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# A Bangladesh Tourism (Museum) Data Integration Model

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Database Systems Project

Scrum Level-01

Scrum Group-11

Report submitted September 6, 2026

A project submitted to Prof. Dr. Rudra Pratap Deb Nath, Department of Computer Science and Engineering, University of Chittagong (CU) in partial fulfillment of the requirements for the Database Systems Lab course. The project is not submitted to any other organization simultaneously. The course teacher holds the copyright. The project cannot be distributed or commercialized without their formal permission.

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# Preface

This report presents the work carried out by Group-11 at the Department of Computer Science and Engineering, University of Chittagong as part of the Database Systems Lab course. The project focuses on the Tourism (Museum) sector assigned to Group-11 as part of the MODELBD course initiative.

The course project differs from a conventional single-team database assignment because the class contributes to a shared large-scale database initiative called MODELBD. The project follows a Scrum of Scrums structure in which different groups work on different sectors while maintaining common conventions where integration requires them. Group-11 focuses on museum information in Bangladesh and develops the data foundation for a tourism-oriented museum database.

We identify museums across Bangladesh, collect museum-level information, investigate museum-specific artifacts where sufficiently detailed public information is available, record source references and images, organize the collected information into structured tables, and derive the Entity-Relationship model from the collected data. The conceptual and logical design has now been followed by an implemented MySQL database, completed ETL loading, and SQL validation evidence.

The report emphasizes the collective work of the team. The members contribute to source discovery, museum research, artifact research, image collection, data organization, modeling, review, and documentation as interconnected parts of the same workflow.

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# Abstract

Bangladesh hosts numerous museums that preserve archaeological, historical, artistic, scientific, military, ethnological, and cultural materials, yet detailed museum and artifact information remains fragmented across heterogeneous online sources. This project addresses the need for a structured and queryable representation of museum information for tourism, while also preserving detailed artifact information that can support students, researchers, and museum visitors.

We identify museums across Bangladesh and collect museum-level information such as location, category, ownership, galleries, opening information, entry fees, contact information, and source references. Where sufficiently detailed public information is available, we also collect artifact-level information including artifact type, material, period, finding place, donor or credit, descriptions, images, and source links. The collected information is verified and organized before we derive an Entity-Relationship model and a relational schema from the real data.

The current dataset contains 63 museum records and 3,505 artifact records. Artifact records currently occur for 34 of the 63 museums, while 89 sources are represented in the broader research source analysis, while 74 source rows are loaded into the current relational source table. Artifact records currently occur for 34 of the 63 museums; the 63-museum master list is therefore broader than current artifact-level coverage. The conceptual design has been updated to reflect the normalized artifact relationships, and the relational implementation contains the geographic, museum, gallery, artifact, donor, classification, material, image, visiting-information, contact, phone, and source relations. MySQL Workbench implementation, data loading, and SQL query testing have now been completed and validated.

The project's significance lies in bringing fragmented museum and artifact information into a structured database design that can support retrieval, filtering, joins, and aggregation for tourism-oriented use cases. Its main limitations are incomplete artifact-level coverage for many museums and unavailable values such as images or precise dimensions in some records; the current database implementation and validation have been completed. Future work therefore focuses on performance evaluation, source maintenance, application development, and gradual extension of artifact coverage.

# 1 Introduction

The objective of the Database Systems Lab course is to develop a database solution by applying the theories, methodologies, tools, and technologies learned throughout the course. This document reports the work of Group-11, assigned to the Tourism (Museum) sector of the shared MODELBD initiative. The report describes how we identify the domain, discover and collect data, organize and verify the collected information, derive a conceptual model from the real-world data, and prepare the resulting design for relational implementation.

The Tourism (Museum) sector is selected as a focused representation of cultural tourism in Bangladesh. The project is not limited to maintaining a list of museum names. It aims to connect museum-level information with geographical hierarchy, galleries, artifact-level information, visiting information, contact information, and source provenance so that the database can support a broad range of future information-retrieval requirements. The primary perspective is tourism: a visitor should be able to identify a museum, understand where it is located, determine how it can be visited, and explore what kinds of collections it contains. At the same time, the depth of artifact information makes the dataset useful for students, researchers, and other users interested in cultural heritage.

The work is carried out as part of a shared sector-based database initiative. This makes consistency important beyond the immediate museum dataset. Geographic concepts such as Division, District, and City are represented explicitly so that the Tourism sector can remain compatible with the common data structures used by the wider initiative. The project therefore combines domain research with database design: the real information collected from museums and artifacts determines the entities, attributes, keys, and relationships that the database needs to represent.

## 1.1 Background and Motivation

Bangladesh carries a cultural heritage that spans more than two millennia and is preserved across a wide range of museums. Large national institutions, regional research museums, heritage-site museums, converted historical residences, and specialized museums together preserve archaeological objects, sculptures, terracotta, coins, manuscripts, textiles, paintings, weapons, scientific specimens, historical documents, and many other forms of cultural material. These collections are valuable not only as museum holdings but also as resources for cultural tourism, education, historical research, and heritage preservation.

The breadth of this landscape creates a practical information-management challenge. A tourist may need to know where a museum is located, its category, opening hours, entry fee, contact information, and available galleries before planning a visit. A researcher may need to identify artifacts by material, period, finding place, type, or museum. A museum visitor may want to explore the collection of a particular institution. These requirements involve information that is usually distributed across different sources rather than maintained in one consistent structure.

General reference platforms such as Wikipedia and Banglapedia provide useful museum-level information, but they do not consistently provide the deeper structured details needed for systematic querying. Official museum pages, government portals, travel guides, news reports, academic repositories, digital archives, and other sources contain additional information, but the sources differ in format, depth, terminology, and completeness. Artifact-level information is particularly difficult to collect because many museums do not publish detailed public catalogues or photographs for individual artifacts.

The motivation for the project therefore comes from the need to transform fragmented museum and artifact information into a unified relational structure. Instead of treating each source as an isolated page, we organize the information around connected entities. This approach creates a foundation for questions such as which museums exist in a district, what artifacts are associated with a museum or gallery, which artifacts are made of a particular material, which museums have particular opening or fee arrangements, and which source supports a particular museum record.

The project is also motivated by the potential for future tourism-oriented applications. The database can serve as a structured foundation for a future website or application through which tourists can discover museums and their collections. At the same time, the detailed artifact records can support academic exploration and heritage-related research. The database therefore treats tourism as the primary use perspective while preserving a level of artifact detail that allows broader cultural and research use.

## 1.2 Problem Statement

The central problem this project addresses is the absence of a structured, queryable database that links museum-level details, such as location, ownership, category, opening hours, and entry fee, with artifact-level details, such as artifact type, material, period, finding place, and images, for museums across Bangladesh. More specifically, we aim to:

1. Integrate museum-level data including location, ownership, category, galleries, opening

hours, entry fees, and contact information with artifact-level data including type, material, period, finding place, donor credit, and images in a single relational schema.

2. Establish a unified schema that captures the geographic hierarchy of Division, District, and City so that the Tourism sector can be consistently integrated with the other sectors of the shared database initiative.
3. Identify and document reliable public sources for every museum and artifact record so that the dataset remains verifiable and extensible.
4. Provide the structural foundation for query capabilities—retrieval, filtering, joins, and aggregation—once the schema is implemented, so that stakeholders can browse museums by district, compare entry fees, or search artifacts by material or period.

Solving this problem in practice, however, means confronting a set of specific, recurring obstacles that surface throughout the data collection process:

1. **Data source unavailability.** Comprehensive lists of Bangladeshi museums exist, but detailed, structured information about any individual museum is often unavailable from a single authoritative source, forcing us to cross-reference multiple sources per museum.
2. **Inconsistent and unstandardized formats.** The identified sources span encyclopedic platforms, official museum websites, government portals, news media, academic repositories, digital archives, and travel guides, each describing details such as opening hours, entry fees, and contact information in its own format, requiring manual reconciliation before the data can be entered consistently.
3. **Missing or hard-to-source artifact images.** With 3,505 artifacts collected across 63 museums, locating a reliable image for each individual artifact proves the single most time-consuming part of the process, since very few museums publish artifact-level photographs online.
4. **Missing or incomplete attribute values.** Fields such as period dating, finding place, and precise dimensions are frequently absent from source material or given only as unstructured free text, requiring us to record what is verifiably available rather than infer missing values.
5. **Duplicate and conflicting records.** Where multiple sources describe the same museum or artifact, the details reported sometimes conflict, requiring us to cross-check and resolve discrepancies before a record is accepted into the dataset.
6. **Scale of artifact-level documentation.** A single museum can hold well over a thousand artifacts, making exhaustive, museum-by-museum documentation within the timeframe of a single course project infeasible, and requiring us to prioritize coverage and data quality over completeness.

Technically, the system encompasses entity types including Museum, Artifact, Gallery, Division, District, City, Owner, Category, Material, Artifact Type, Image, Contact, Entry Fee, Opening Hours, Closed Days, and Source, connected through relationships such as exhibits, made of, owned by, classified as, and provides. Together, these structures capture geographic hierarchy, ownership, classification, gallery structure, and the administrative detail of each museum.

### **1.3 System Definition**

A computerized system used to store, organize, and retrieve information about museums and artifacts across Bangladesh by capturing museum profiles, artifact records, geographic hierarchy, and administrative details through a relational database. The system supports queries such as locating museums by district, retrieving artifacts by material or period, and comparing entry fees and opening hours across institutions, and is designed to remain extensible as additional museums and artifacts are documented in future work.

### **1.4 Comparison of Traditional Museum Information Management and the Proposed Database System**

During our data collection, we found that museum information is distributed across different websites, documents, travel resources, and other online sources. A visitor who wants to find a museum's location, opening hours, entry fee, or important artifacts may therefore need to consult several sources separately. Our proposed database brings the verified information into a structured and searchable relational model. The comparison below summarizes the practical difference.

Table 2: Traditional museum information management versus the proposed database system

| Aspect               | Traditional / Existing Approach                                              | Proposed Tourism (Museum) Database                                                          |
|----------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Data source          | Information is scattered across websites, PDFs, articles, and other sources. | Verified sources are recorded systematically with source references.                        |
| Museum search        | Users often search several websites separately.                              | Museum information can be retrieved from a single structured database.                      |
| Artifact information | Artifact details are often scattered or incomplete.                          | Artifact records are organized separately and linked to museums.                            |
| Data structure       | Sources use different formats and naming conventions.                        | Attributes and values are standardized before modelling.                                    |
| Source tracking      | The original evidence can be difficult to trace.                             | Source links are preserved for traceability and verification.                               |
| Tourist use          | Repeated web searching is often required for basic visit information.        | Location, opening information, entry fees, contacts, and artifacts can be queried together. |
| Research use         | Comparing detailed artifact information across museums is difficult.         | Structured artifact records support filtering, joins, and future analysis.                  |
| Data quality         | Duplicate, missing, or inconsistent records may occur across sources.        | Records are cross-checked, cleaned, and validated before database implementation.           |
| Future analysis      | Manual collection and comparison are time-consuming.                         | SQL queries can support search, comparison, counting, and aggregation after implementation. |

This comparison reflects the problem we encountered rather than a purely theoretical difference. The main purpose of the database is tourism-oriented information retrieval, while the detailed artifact structure also makes the collected dataset useful for students, researchers, and museum visitors.

## 1.5 System Development Process

We follow a Scrum-based development process within the shared MODELBD structure. The actual work begins when the broader Tourism project is assigned and the mentor directs us to focus on museums. We first identify museums across Bangladesh and collect the information that can be found online about each museum. We then investigate the artifacts associated with individual museums and collect artifact information and images where sufficiently reliable public information is available.

The collected information is organized into rows and columns in shared spreadsheet tables. We then examine the structure of the real data to determine what entity types, attributes, keys, and

relationships are required. The Entity-Relationship model is derived from this analysis rather than being created independently of the collected information. After the conceptual model is reviewed, we prepare the relational schema, check the design against normalization principles, and implement the resulting database in MySQL Workbench. ETL loading, query development, testing, and final validation are documented in the implementation section.

Figure 1 summarizes this process as twelve stages grouped into four phases. The Initiation phase establishes the scope. Data Acquisition covers source identification, collection, and verification. Database Modelling converts the collected information into an Entity-Relationship design, relational schema, and normalized structure. The final Implementation and Testing phase covers MySQL Workbench schema creation, ETL loading, SQL query execution, and validation, which are documented in the completed implementation evidence below.

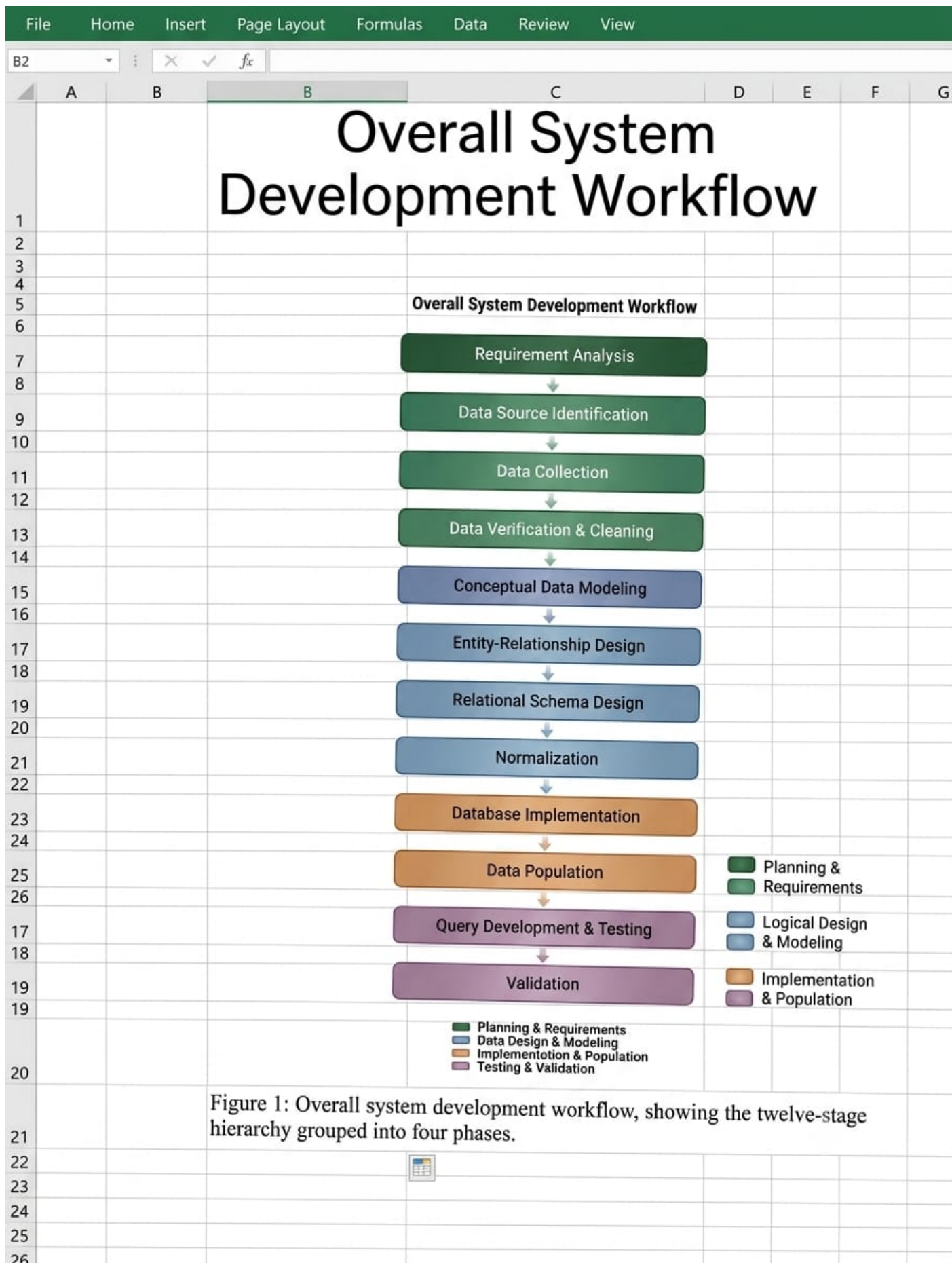


Figure 1: Overall system development workflow, showing the twelve-stage hierarchy grouped into four phases

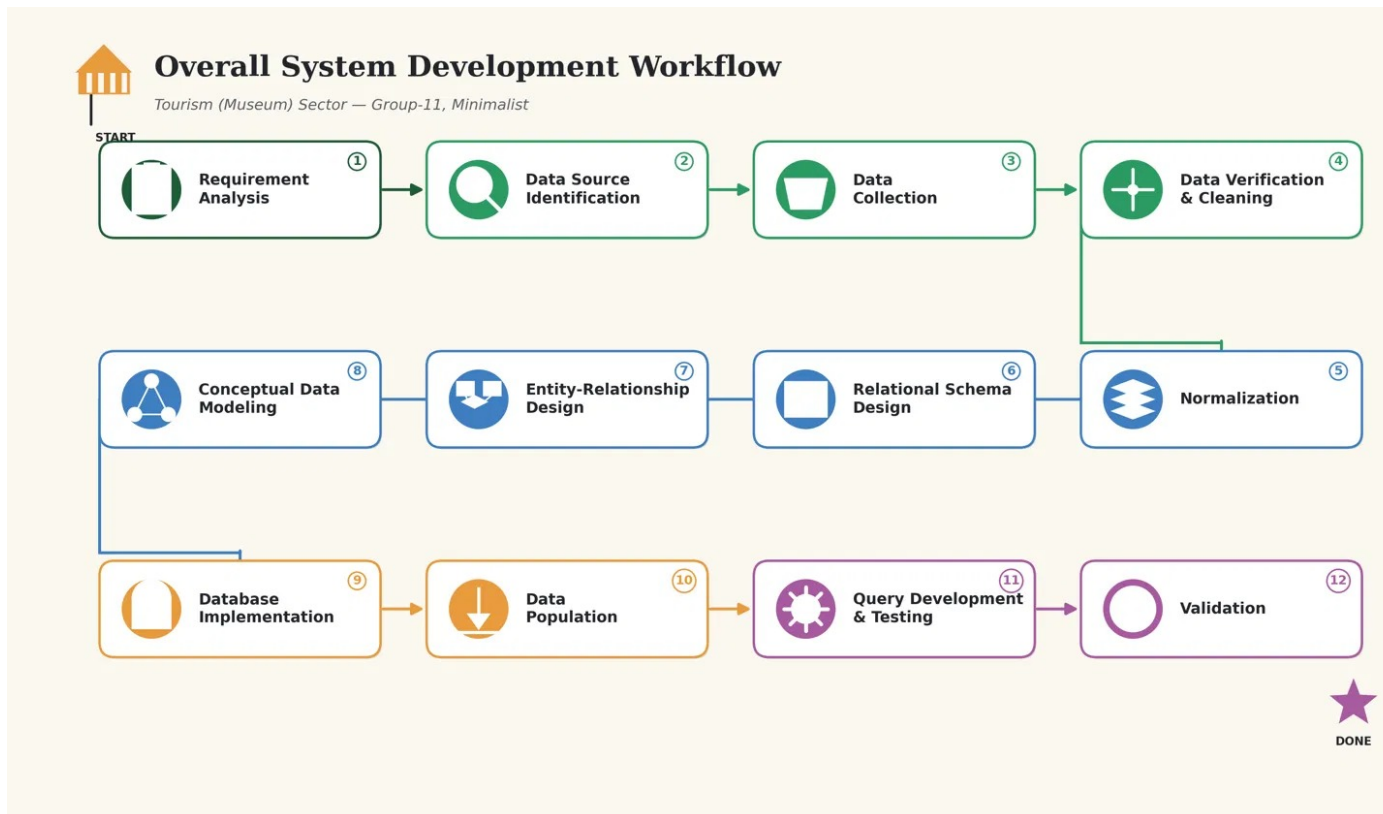


Figure 2: Data Flow Diagram (DFD) Level 1 for the Tourism (Museum) database system

Figure 2 presents the Level 1 data flow at a practical level. External sources such as government websites, official museum pages, online references, reports, and other field or supporting sources provide the raw information. The internal processes then move the information through source collection, extraction, cleaning and standardization, entity mapping, consolidation, validation, modelling, and preparation for database loading. The diagram separates the prepared museum and artifact data from the MySQL database and is complemented by the implemented ETL workflow documented in Section 3.9.

## 1.6 Organization

The remainder of this document follows the structure supplied for the Database Systems Lab project report. Chapter 2 describes team organization, roles, tools, shared data resources, the sprint backlog, and sprint structure. Chapter 3 documents Sprint 1, including source discovery, data acquisition, challenges, the Entity-Relationship model, entities, relationships, data population, and the implemented Extract, Transform, Load pipeline. Chapter 4 describes deployment and maintenance. Chapter 5 discusses intra-group and inter-group collaboration, good memories, challenges, and lessons learned. Chapter 6 reflects on course pedagogy. Chapter 7 presents the logical model and relational schema. Finally, Chapter 9 concludes the report and identifies limitations and future work.

## 2 Project Structure and Management

Group-11 organizes the project using a lightweight Scrum-inspired structure suited to a four-member team. The work is treated as a shared responsibility rather than as isolated individual assignments. Trello Board provides task visibility, Google Drive and Google Sheets provide shared data storage, Whats App supports daily communication, and Overleaf supports collaborative documentation. The management structure keeps the progress of museum research, artifact research, data organization, modeling, and database implementation visible to the whole team.

### 2.1 Team Organization and Roles

Table 3 summarizes the roles used in the project. The responsibilities overlap intentionally because museum research, artifact research, source verification, data organization, modeling, and documentation are connected stages of one workflow. The table therefore describes primary responsibilities rather than exclusive ownership of isolated tasks.

Table 3: Team roles and responsibilities

| Role          | Name                                 | Primary responsibilities                                                                                                                                                                  |
|---------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scrum Manager | Sadia Binte Bokthear Ekra (24701065) | Source collection, museum list preparation, museum-wise artifact data collection, artifact image collection, Entity-Relationship modelling, task coordination, and documentation support. |
| Developer     | Sabina Sultana (24701015)            | Museum list preparation, source collection, Entity-Relationship modelling, schema preparation, and documentation support.                                                                 |
| Developer     | Fatema Tuj Johra (24701011)          | Museum and artifact data collection, artifact image collection, data organization, and preparation for later testing.                                                                     |
| Developer     | Nowshin Ara Nargish (24701030)       | Museum research, museum-wise artifact data collection, source verification, and preparation of database records.                                                                          |

Our team structure is important because the project requires repeated cross-checking. A museum record can change after another member finds a better source, and artifact details can require additional verification. Shared responsibility therefore supports consistency and reduces dependence on one member's local copy of the data.

### 2.2 Tools and Technologies

The project uses a small set of tools with clearly separated purposes. Trello is used for task tracking and work-stage visibility. GitHub is used as the project repository for version-controlled database, ETL, documentation, and supporting files. Google Drive and Google Sheets provide the shared research environment for museum records, artifact records, source information, and images. WhatsApp supports quick coordination and clarification. Overleaf is used for collaborative report preparation. MySQL Workbench is the implemented environment

for creating the relational database, loading the prepared data, and testing Structured Query Language queries. We also use ChatGPT, Claude, and Copilot as AI-assisted tools for source discovery, repetitive data preparation, checking possible anomalies, and documentation support, with final information verified against the original sources.

Table 4: Tools and their roles in the project

| <b>Tool</b>                  | <b>Purpose</b>                                                                                                                                                                                                                                                    |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trello                       | Task tracking, sprint backlog organization, progress visibility, and work-stage labels.                                                                                                                                                                           |
| GitHub                       | Version-controlled project repository for the database, ETL scripts, documentation, and supporting files. <a href="https://github.com/FatematujJohra/Bangladesh-Museum-Tourism-Database">https://github.com/FatematujJohra/Bangladesh-Museum-Tourism-Database</a> |
| Google Drive / Google Sheets | Shared museum data, artifact data, source records, images, and research materials.                                                                                                                                                                                |
| WhatsApp                     | Daily team communication, quick coordination, task updates, and clarification.                                                                                                                                                                                    |
| Overleaf                     | Collaborative LaTeX report preparation and revision.                                                                                                                                                                                                              |
| MySQL Workbench              | Database implementation, schema creation, ETL data loading, and Structured Query Language query testing.                                                                                                                                                          |
| ChatGPT                      | AI-assisted source discovery, source-based information extraction, data organization, cleaning checks, and documentation support.                                                                                                                                 |
| Claude                       | AI-assisted deep-search, source discovery, structured data extraction, and repetitive data-preparation support.                                                                                                                                                   |
| GitHub Copilot               | AI-assisted coding and database/query preparation support during the implementation stage.                                                                                                                                                                        |
| Python                       | A Python script extracted and formatted project data into CSV files. These files were then programmatically parsed and ingested via database connectors to automatically populate tables in MySQL Workbench.                                                      |

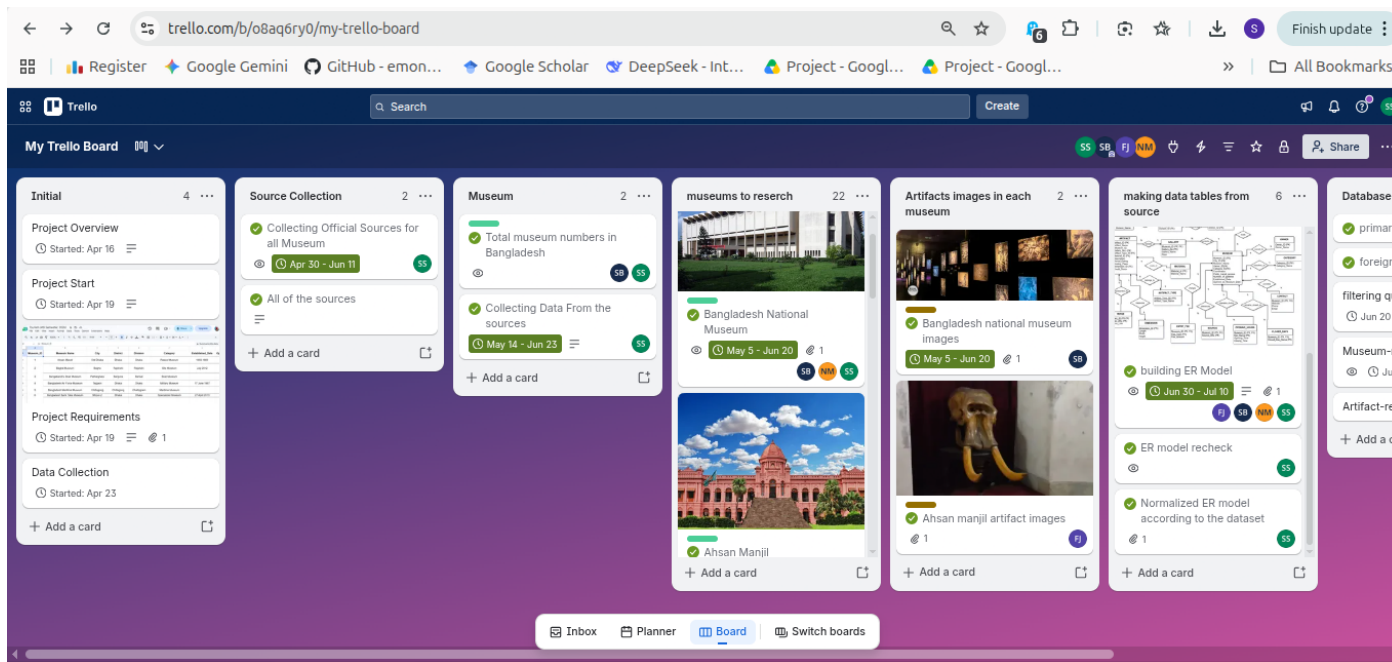


Figure 3: Group-11 Trello board showing the list structure used to track project work  
<https://trello.com/invite/b/6a368fa8b80c8050f7d19104/ATTI1cf8e8d855a08aac056f5fd03997b406C15BEF96/my-trello-board>

Figure 3 shows the project workflow from initial setup through museum research, artifact and image collection, structured data preparation, and database implementation. The Database Creation stage has since been completed: the relational schema was implemented in MySQL Workbench, the normalized CSV data were loaded, and SQL queries were executed for record-count, relationship, analytical, and integrity validation. The board documents the workflow, while the completed implementation is evidenced in the ETL and SQL sections of this report.

**GitHub repository link.** The project repository is maintained at <https://github.com/FatematujJohra/Bangladesh-Museum-Tourism-Database>. The GitHub repository is used to keep the database schema, prepared CSV data, ETL loader, SQL validation/finding queries, documentation, and execution evidence together under version control.

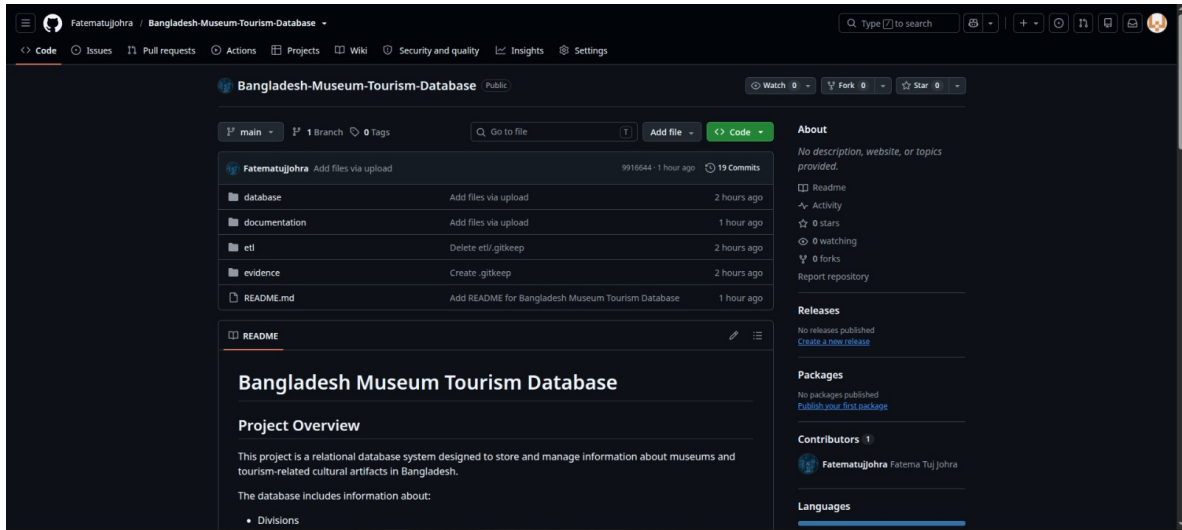


Figure 4: GitHub repository and README evidence for the Bangladesh Museum Tourism Database project

Figure 4 shows the actual project repository page, including the repository name and its README.md entry point. The README begins with the project title “Bangladesh Museum Tourism Database” and a project overview describing the work as a relational database system for museums and tourism-related cultural artifacts in Bangladesh. The visible repository layout also separates the database, etl, documentation, and evidence areas.

### 2.2.1 Repository README and Reproducibility

The repository README and folder structure provide a concise map of how the implementation is organized. The database/ directory contains the SQL schema, CSV data, and query scripts; etl/ contains the Python loader and its dependency/implementation notes; documentation/ contains the ER, ETL, and finding documents; and evidence/ contains execution screenshots. This organization mirrors the actual implementation workflow described later in the report: prepare the data, create the schema, run the ETL loader, validate the database, and retain the resulting evidence.

The GitHub link is intentionally placed immediately after the Trello evidence so that project management evidence and version-control evidence can be viewed together. GitHub is not treated as a source of museum facts; it is the repository used to maintain the project’s own implementation artifacts.

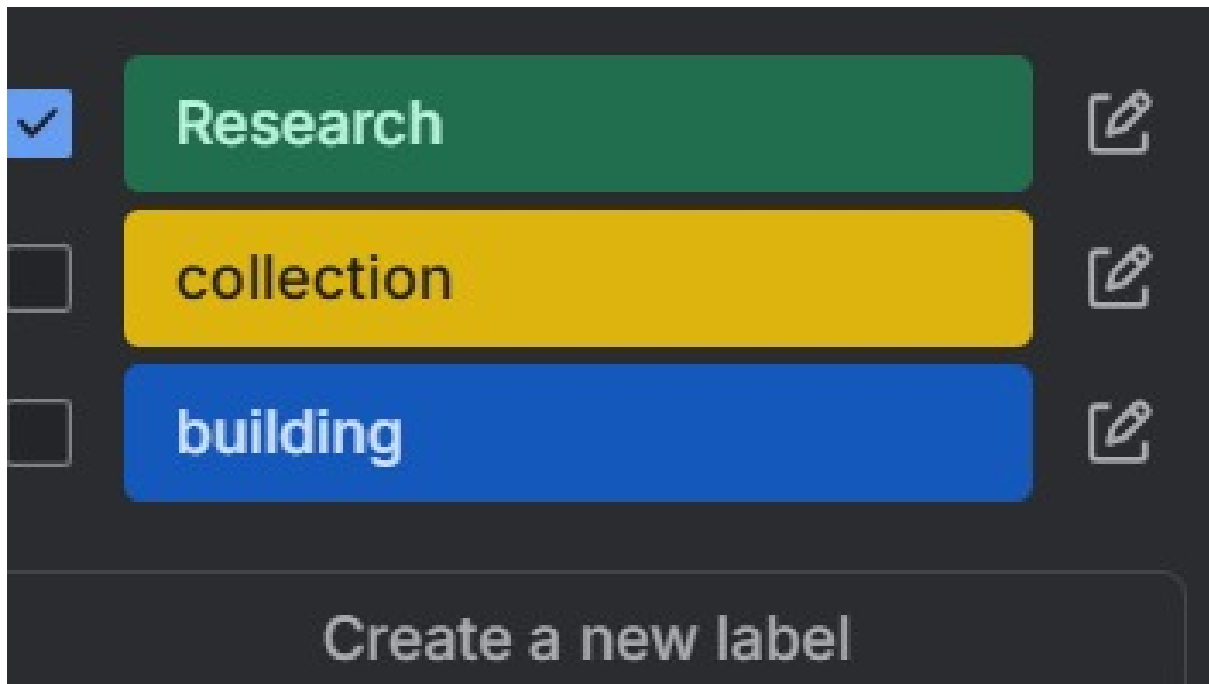


Figure 5: Trello labels used to distinguish Research, collection, and building work stages

Figure 5 shows the three-color label scheme that we layer on top of the list structure in Figure 3 to mark what kind of work a card represents, independent of which list it currently sits in. The green label reads Research and is applied to cards where a member is still identifying a museum, verifying its existence, or locating its official sources—the earliest and most exploratory stage of work on any given museum. The yellow label reads collection and marks cards where we have already confirmed a museum is a valid target and is now actively gathering artifact images, opening hours, entry fees, and other structured details from the sources identified during the research stage. The blue label reads building and marks the final stage, where verified museum and artifact information is assembled into the structured spreadsheet tables that ultimately feed the relational database; the “museum data table” and “Artifacts data table” cards under Making Data Tables from Source in Figure 3 both carry this blue label. Because a card can move through green, then yellow, then blue as work on its museum progresses, the label color gives every member an immediate visual answer to “how far along is this museum” without needing to open the card itself—a lightweight substitute for a full Scrum burndown chart that suited a four-person team working asynchronously over WhatsApp.

## 2.3 Data Sources and Shared Repository

All source records, museum-level data, and artifact-level data are maintained in a shared Google Sheets workbook. The master source information records the references used during collection and verification, while museum-specific sheets organize deeper artifact information where sufficiently detailed public information is available. Keeping we on the same shared workbook is important because individual records often require correction after cross-checking against a second or third source. Drive Link-<https://drive.google.com/drive/folders/178TY1-m9VOAUsukBcBaYnCeDMXAuV-zE?usp=drive.link>

GitHub Link-<https://github.com/FatematujJohra/Bangladesh-Museum-Tourism-Database>

Table 5: Primary shared data sources for the Tourism (Museum) sector

| Sheet / Resource                             | Link                                                                                                                                                                                                                                                        |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Master source list                           | <a href="https://docs.google.com/spreadsheets/d/1vCkqwgUvZJFsK4h72KsDRgJgmmoyfraxLZV-LojRfY/edit?gid=1799459655#gid=1799459655">https://docs.google.com/spreadsheets/d/1vCkqwgUvZJFsK4h72KsDRgJgmmoyfraxLZV-LojRfY/edit?gid=1799459655#gid=1799459655</a>   |
| Museum list / museum data table (63 museums) | <a href="https://docs.google.com/spreadsheets/d/13iet5oXoujhXu2CpOcdGDiqGu3Q9_qW3wnuSWDmhbzQ/edit?gid=1769566929#gid=1769566929">https://docs.google.com/spreadsheets/d/13iet5oXoujhXu2CpOcdGDiqGu3Q9_qW3wnuSWDmhbzQ/edit?gid=1769566929#gid=1769566929</a> |
| Artifacts List                               | <a href="https://docs.google.com/spreadsheets/d/1AotKvFkbbonyPMdQI0n5xFv_wu9Qdkqy/edit?gid=1946217238#gid=1946217238">https://docs.google.com/spreadsheets/d/1AotKvFkbbonyPMdQI0n5xFv_wu9Qdkqy/edit?gid=1946217238#gid=1946217238</a>                       |

Table 5 provides direct access to the shared master source list and museum data table. The workbook is the working evidence base for the project: source references, museum records, artifact research, and museum-specific sheets remain traceable rather than being treated as disconnected notes.

## 2.4 Sprint Backlog

The sprint backlog is organized around the work we planned for the current project stage and the remaining implementation work. The completed portion covers project scoping, museum identification, data collection, source verification, data organization, conceptual modelling, and relational schema preparation. Database implementation, data loading, and query testing have now been completed.

Table 6: Detailed sprint backlog for the Tourism (Museum) database

| ID    | Backlog item                                                                            | Priority | Status    |
|-------|-----------------------------------------------------------------------------------------|----------|-----------|
| PB-01 | Define the Tourism project scope and confirm the Museum focus.                          | High     | Completed |
| PB-02 | Define project objectives and identify the primary and secondary users.                 | High     | Completed |
| PB-03 | Identify museums across Bangladesh and prepare the master museum list.                  | High     | Completed |
| PB-04 | Collect museum location, division, district, city, category, and ownership information. | High     | Completed |
| PB-05 | Collect gallery, opening-hour, closed-day, entry-fee, contact, and website information. | High     | Completed |

| <b>ID</b> | <b>Backlog item</b>                                                                                                          | <b>Priority</b> | <b>Status</b> |
|-----------|------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------|
| PB-06     | Collect and maintain source records for museum information and verification.                                                 | High            | Completed     |
| PB-07     | Research museum-specific artifact collections where reliable public information is available.                                | High            | Completed     |
| PB-08     | Collect artifact names, types, materials, dating, finding places, dimensions, descriptions, and donor or credit information. | High            | Completed     |
| PB-09     | Collect and verify available artifact images and image-source links.                                                         | High            | Completed     |
| PB-10     | Organize museum and artifact information into structured rows and columns.                                                   | High            | Completed     |
| PB-11     | Re-examine sources, cross-check conflicting values, identify missing information, and avoid unsupported estimates.           | High            | Completed     |
| PB-12     | Identify entities and attributes from the structure of the collected real-world data.                                        | High            | Completed     |
| PB-13     | Identify primary keys, foreign-key requirements, and relationships among the entities.                                       | High            | Completed     |
| PB-14     | Build and review the Entity-Relationship model.                                                                              | High            | Completed     |
| PB-15     | Convert the conceptual model into the relational schema and check normalization.                                             | High            | Completed     |
| PB-16     | Implement the relational schema in MySQL Workbench with primary keys, foreign keys, and integrity constraints.               | High            | Completed     |
| PB-17     | Load the prepared museum, artifact, and supporting records into the implemented database.                                    | High            | Completed     |
| PB-18     | Develop retrieval, filtering, join, and aggregation queries for tourism and research use cases.                              | Medium          | Completed     |
| PB-19     | Validate query results, relationships, constraints, and data consistency.                                                    | Medium          | Completed     |
| PB-20     | Prepare the final documentation, evidence figures, and implementation results.                                               | Medium          | Completed     |

The backlog reflects the project’s actual development state. The research, collection, validation, modelling, and schema-preparation items are supported by our shared data and Trello records. The MySQL Workbench, data-loading, query-testing, and validation items are now completed and supported by execution evidence included in the ETL section.

## 2.5 Representative User Stories

The sprint backlog can also be interpreted through representative user stories. These stories clarify why the collected data matters to different stakeholders and keep the database design connected to practical use cases.

Table 7: Representative user stories underlying the sprint backlog

| ID    | User Story                                                                                                          | Priority |
|-------|---------------------------------------------------------------------------------------------------------------------|----------|
| PB-01 | As a curator, I want to access integrated artifact records so that I can manage collections effectively.            | High     |
| PB-02 | As a heritage researcher, I want to search artifacts by category, period, and museum to support historical studies. | High     |
| PB-03 | As a tourist, I want to view museum information, opening hours, and entry fees to plan a visit.                     | Medium   |
| PB-04 | As a museum administrator, I want to compare my institution’s data with peer museums in the same district.          | Medium   |
| PB-05 | As an administrator, I want to update artifact and source information as better records become available.           | Low      |

Table 7 connects the backlog to actual stakeholders. Although tourists are the primary target users of the Tourism (Museum) sector, the detailed artifact structure supports research and museum-management use cases as well.

## 2.6 Sprint Structure

The project progresses through incremental stages. The first sprint establishes the data foundation through source identification, museum identification, museum-level collection, artifact research, source verification, and data organization. The modelling stage then converts the collected information into an Entity-Relationship model and relational schema. The implementation and testing tasks were completed in the current increment, while future work focuses on maintenance, extension of artifact coverage, and application development.

## 2.3 SPRINT STRUCTURE

The project is executed in iterative sprints with regular reviews and continuous integration.

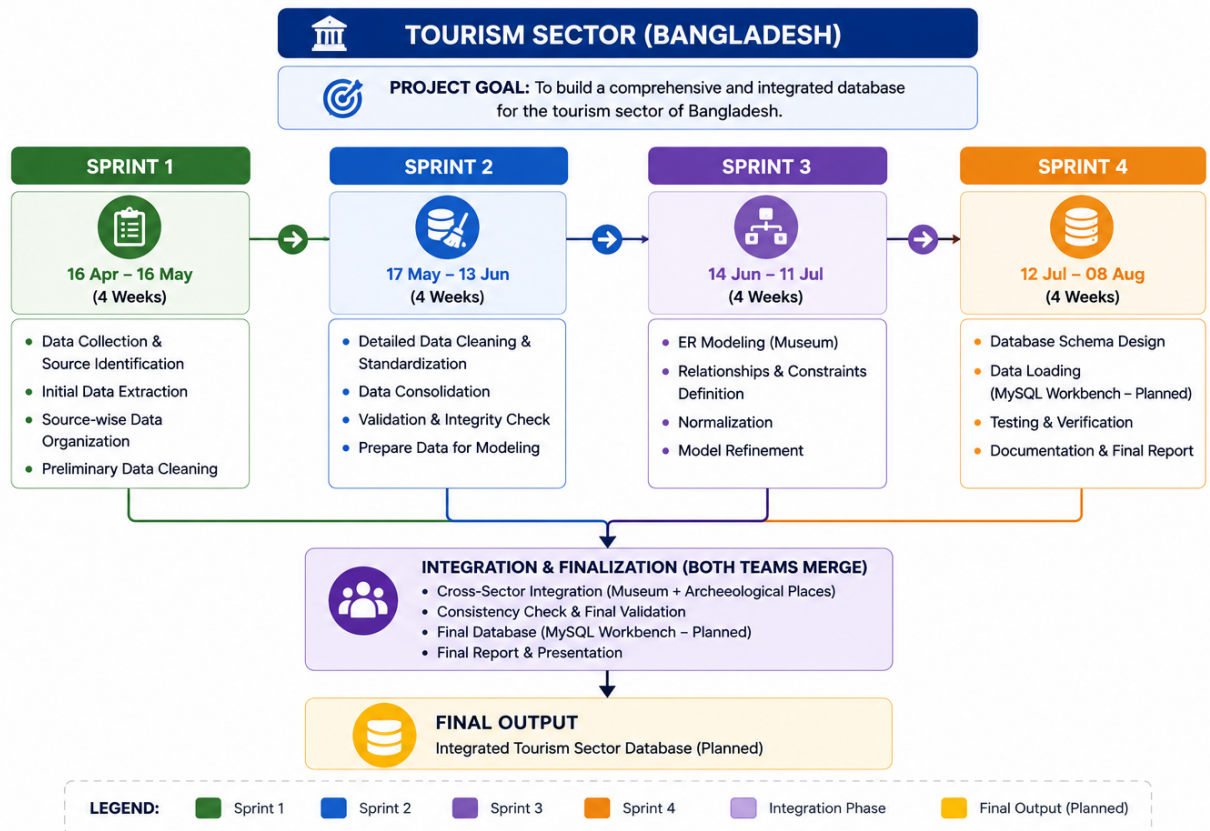


Figure 6: Sprint structure of the Tourism (Museum) project, including the final integration phase

Figure 6 shows how the project is divided into four iterative sprints followed by an integration and finalization phase. Sprint 1 establishes the data foundation through source identification and collection. Sprint 2 focuses on cleaning, standardization, consolidation, and validation. Sprint 3 concentrates on the Entity-Relationship model, relationships, constraints, normalization, and model refinement. Sprint 4 covers relational implementation, data loading, testing, and final documentation. The final integration phase is where the sector-level work is brought together into the common Tourism database, followed by consistency checking and final validation. The dates shown in the figure are the working sprint schedule used by our project: 16 April–16 May, 17 May–13 June, 14 June–11 July, and 12 July–08 August, respectively.

## 3 Sprint 1

Sprint 1 establishes the data foundation of the Tourism (Museum) database. The sprint covers requirement understanding, source discovery, museum identification, museum-level data collection, artifact research, source verification, data organization, and the early conceptual modelling that follows from the collected data. We treat the shared spreadsheet as the working data environment during this stage.

### 3.1 Sprint Overview

The sprint begins after the Tourism domain is assigned and the mentor directs us to focus on museums. We first identify museums across Bangladesh and prepares a museum list. It then collects available information about each museum from online sources. After that, we investigate artifacts museum by museum, searches for artifact descriptions and images, and organizes the information into structured rows and columns. The resulting structure becomes the basis for identifying entities and relationships.

### 3.2 Sprint Backlog

The Sprint 1 backlog concentrates on the data foundation. The main tasks are summarized below.

Table 8: Sprint 1 backlog

| ID    | Sprint task                                                                                                          | Status    |
|-------|----------------------------------------------------------------------------------------------------------------------|-----------|
| S1-01 | Identify the museum focus within the Tourism sector and define the information scope.                                | Completed |
| S1-02 | Identify museums across Bangladesh and prepare the master museum list.                                               | Completed |
| S1-03 | Collect museum-level information from multiple sources.                                                              | Completed |
| S1-04 | Investigate museum-specific artifacts and collect available artifact details and images.                             | Completed |
| S1-05 | Organize the collected information into structured spreadsheet rows and columns.                                     | Completed |
| S1-06 | Verify sources, cross-check conflicting information, and preserve missing values where verification is not possible. | Completed |
| S1-07 | Identify entities, attributes, keys, and relationships from the structured data.                                     | Completed |
| S1-08 | Prepare and review the Entity-Relationship model.                                                                    | Completed |

### 3.3 Data Discovery Process

We begin Sprint 1 by identifying reliable sources of information about museums and artifacts across Bangladesh. The search covers encyclopedic and reference platforms, official museum

sites, government portals, travel guides, academic repositories, digital archives, news media, and social media. The source records are maintained with source links and related reference information so that the collected data can be traced back to its evidence.

### Data source hierarchy: category-wise and source-wise contribution (n = 89 identified sources)

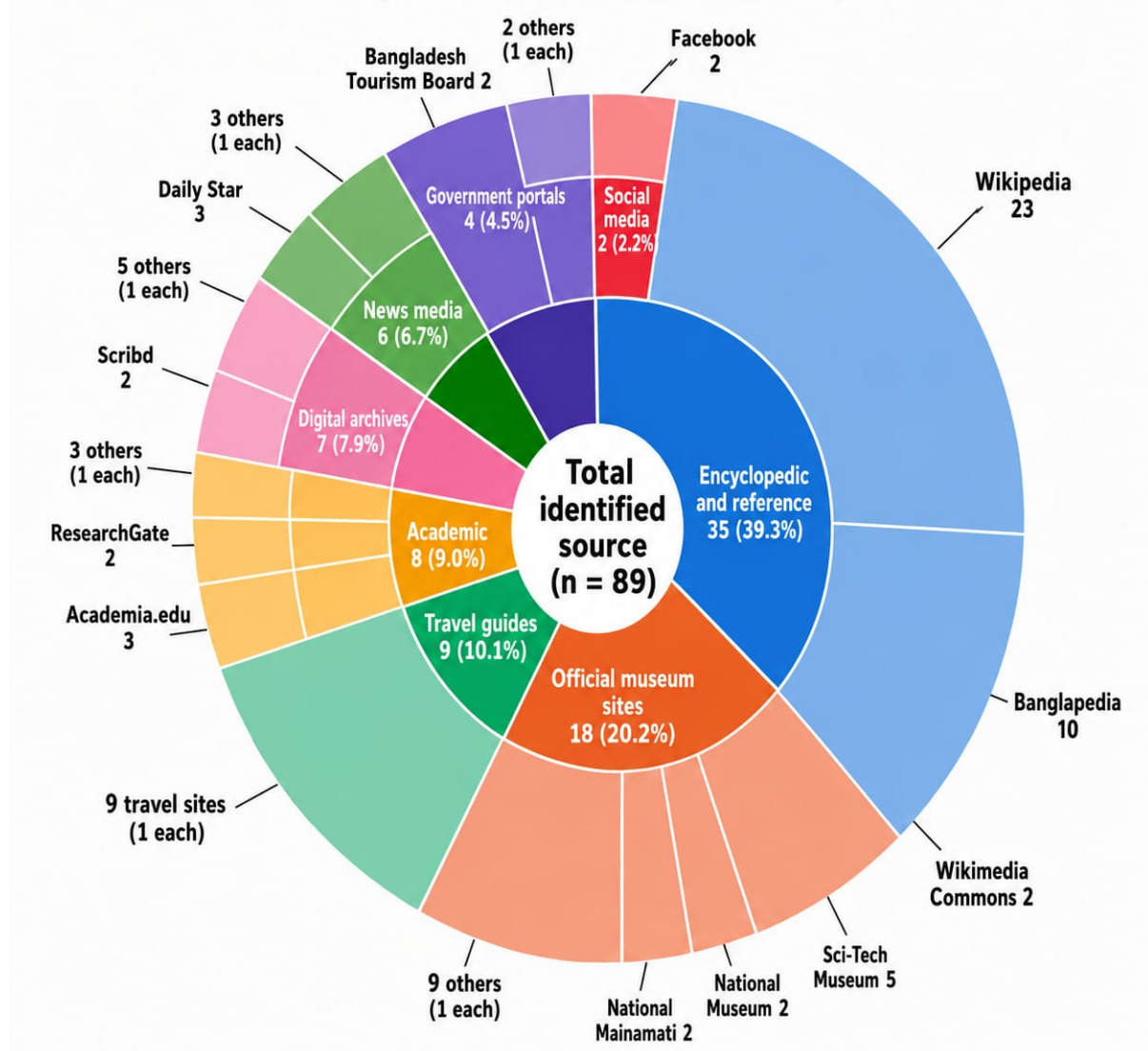


Figure 7: Hierarchical breakdown of the 89 identified data sources by category and individual source contribution

Figure 7 presents the distribution of the 89 identified sources by category and by individual source contribution. The inner ring groups sources into broad categories, while the outer ring shows the individual sources within those categories. Encyclopedic and reference sources form the largest category with 35 sources (39.3%), followed by official museum sites with 18 sources (20.2%). Travel guides contribute 9 sources (10.1%), academic sources contribute 8 (9.0%), digital archives contribute 7 (7.9%), news media contribute 6 (6.7%), government portals contribute 4 (4.5%), and social media contributes 2 (2.2%).

Table 9: Sources Used for Museum and Artifact Data Collection

| Source Category                          | No. | Source Examples                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Main Information Collected / Verified                                                              |
|------------------------------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Encyclopedic & Reference Sources         | 35  | <b>Wikipedia</b> – <a href="http://wikipedia.org">wikipedia.org</a><br><b>Banglapedia</b> – <a href="http://en.banglapedia.org">en.banglapedia.org</a><br>Other reference sources identified during research                                                                                                                                                                                                                                                                                                                                                                                                               | Museum discovery, historical background, general descriptions and cross-checking                   |
| Official Museum / Institutional Websites | 18  | <b>Bangladesh National Museum</b> – <a href="http://bangladeshmuseum.gov.bd">bangladeshmuseum.gov.bd</a><br><b>Ahsan Manzil Museum</b> – <a href="http://ahsanmanzil.org.bd">ahsanmanzil.org.bd</a><br><b>Varendra Research Museum</b> – <a href="http://vrmbd.ru.ac.bd">vrmbd.ru.ac.bd</a><br><b>Liberation War Museum</b> – <a href="http://liberationwarmuseumbd.org">liberationwarmuseumbd.org</a><br><b>Bangladesh Folk Art &amp; Crafts Foundation</b> – <a href="http://sonargaonmuseum.gov.bd">sonargaonmuseum.gov.bd</a><br><b>Bangladesh Air Force Museum</b> – <a href="http://bafmuseum.com">bafmuseum.com</a> | Museum identity, location, visiting information, collections, exhibitions and artifact information |
| Travel & Tourism Sources                 | 9   | <b>Bangladesh Tourism Board / Beautiful Bangladesh</b> – <a href="http://beautifulbangladesh.gov.bd">beautifulbangladesh.gov.bd</a><br>Other travel and tourism information websites                                                                                                                                                                                                                                                                                                                                                                                                                                       | Location, attractions, opening information, entry fees and visitor-oriented information            |
| Academic & Research Sources              | 8   | <b>ResearchGate</b> – <a href="http://researchgate.net">researchgate.net</a><br>Academic journal articles<br>Research publications<br>Institutional research pages                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Historical context, artifact interpretation, museum studies and detailed research information      |
| Digital Archives                         | 7   | <b>Liberation War Museum Digital Library</b> – <a href="http://library.liberationwarmuseumbd.org">library.liberationwarmuseumbd.org</a><br><b>Liberation War Museum Photo Archive</b><br><b>Liberation War Museum Oral History Archive</b><br>Museum collection and digital archive pages                                                                                                                                                                                                                                                                                                                                  | Artifact records, photographs, historical documents, oral histories and collection details         |

| Source Category    | No.       | Source Examples                                                                                                                                                                                                                                                                                       | Main Information Collected / Verified                                                                  |
|--------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| News Media         | 6         | <b>Prothom Alo</b> – <a href="http://prothomalo.com">prothomalo.com</a><br><b>The Daily Star</b> – <a href="http://thedailystar.net">thedailystar.net</a><br><b>Dhaka Tribune</b> – <a href="http://dhakatribune.com">dhakatribune.com</a><br>Other newspaper/news sources identified during research | Recent museum information, exhibitions, events, collection-related news and supplementary verification |
| Government Portals | 4         | <b>Ministry of Cultural Affairs</b> – <a href="http://mocca.gov.bd">mocca.gov.bd</a><br>Government cultural and tourism portals<br>Government institutional pages                                                                                                                                     | Official institutional, administrative, ownership and contact information                              |
| Social Media       | 2         | Official/verified museum Facebook pages<br>Official museum social-media accounts                                                                                                                                                                                                                      | Recent photographs, exhibitions, events and supplementary visitor information                          |
| <b>Total</b>       | <b>89</b> | <b>Multiple national and institutional sources – museum-level and artifact-level data collection, verification and update</b>                                                                                                                                                                         |                                                                                                        |

Table 9 describes the broader research source inventory maintained during the collection stage. For reproducibility, the audit below is based specifically on the current v6 ‘source.csv’ relation that is actually loaded into MySQL: 74 source rows across 33 unique domains. The broader research inventory and the loaded relational source table therefore should not be treated as the same count.

The figure demonstrates why source verification is necessary. No single source category provides all of the information required by the database. Reference sources offer broad museum coverage, official sources often provide authoritative institutional information, travel guides contribute visitor-oriented details, and academic or archival sources can provide deeper artifact information. The source diversity therefore becomes a feature of the data collection process rather than an incidental detail.

### 3.3.1 Source Reliability and Provenance Audit

Because the teacher feedback specifically identified the use of third-party sources as a concern, the final dataset was audited by source domain rather than simply described as “verified.” The audit distinguishes primary/first-party institutional sources from institutional/academic supporting sources, secondary/discovery sources, and visual-supporting sources. This classification is intentionally conservative and is based on the organization/domain represented by each URL; it is not a claim that every individual page was independently re-opened or that every statement on a page is authoritative.

The current v6 source.csv contains 74 source rows across 33 unique domains. Of these rows, 28 are classified as Level 1 primary/official or first-party institutional sources, 1 as

Level 2 institutional/academic supporting, 42 as Level 3 secondary/discovery/contextual, and 3 as Level 4 visual supporting. The audit therefore does **not** claim that all museum source rows are primary. Instead, it makes the source hierarchy explicit and identifies where primary-source strengthening is still required.

Table 10: Source hierarchy used for the museum database provenance audit

| Level | Source class                                    | Role in this project                                                                                                                                                                              |
|-------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L1    | Primary / official or first-party institutional | Preferred evidence for museum identity, institutional facts, official collection information, and directly published museum information.                                                          |
| L2    | Institutional / academic supporting             | University, institutional, or research-based material that can provide reliable contextual or research evidence when appropriate.                                                                 |
| L3    | Secondary / discovery / contextual              | Wikipedia, Banglapedia, news, travel sites, blogs, and similar sources. Used for discovery, contextual checking, or supplementary information rather than automatically treated as authoritative. |
| L4    | Visual supporting                               | Image/context sources such as Wikimedia Commons or image repositories. Used primarily for visual evidence, not as the sole authority for factual database attributes.                             |

Table 11: Actual v6 museum-source audit by domain

| Domain                                                                         | Tier | Rows | Museums | Museum IDs |
|--------------------------------------------------------------------------------|------|------|---------|------------|
| <a href="http://ahsanmanzil.org.bd">ahsanmanzil.org.bd</a>                     | L1   | 1    | 1       | 1          |
| <a href="http://bangladeshmuseum.gov.bd">bangladeshmuseum.gov.bd</a>           | L1   | 2    | 1       | 8          |
| <a href="http://bb.org.bd">bb.org.bd</a>                                       | L1   | 1    | 1       | 6          |
| <a href="http://beautifulbangladesh.gov.bd">beautifulbangladesh.gov.bd</a>     | L1   | 1    | 1       | 50         |
| <a href="http://liberationwarmuseumbd.org">liberationwarmuseumbd.org</a>       | L1   | 3    | 1       | 30         |
| <a href="http://madaripurmuseum.org">madaripurmuseum.org</a>                   | L1   | 1    | 1       | 32         |
| <a href="http://mainamati.archaeology.gov.bd">mainamati.archaeology.gov.bd</a> | L1   | 1    | 1       | 34         |

| Domain                                    | Tier | Rows | Museums | Museum IDs                                                           |
|-------------------------------------------|------|------|---------|----------------------------------------------------------------------|
| <a href="#">museum.cu.ac.bd</a>           | L1   | 9    | 1       | 62                                                                   |
| <a href="#">nmst.gov.bd</a>               | L1   | 5    | 1       | 38                                                                   |
| <a href="#">osmanimuseum.org.bd</a>       | L1   | 1    | 1       | 43                                                                   |
| <a href="#">stamforduniversity.edu.bd</a> | L1   | 1    | 1       | 61                                                                   |
| <a href="#">vrmbd.ru.ac.bd</a>            | L1   | 1    | 1       | 55                                                                   |
| <a href="#">ziamuseum.org.bd</a>          | L1   | 1    | 1       | 57                                                                   |
| <a href="#">culture360.asef.org</a>       | L2   | 1    | 1       | 62                                                                   |
| <a href="#">academia.edu</a>              | L3   | 2    | 2       | 34, 61                                                               |
| <a href="#">adarbepari.com</a>            | L3   | 1    | 1       | 58                                                                   |
| <a href="#">alonlytraveler.com</a>        | L3   | 1    | 1       | 23                                                                   |
| <a href="#">bangladesh.com</a>            | L3   | 1    | 1       | 54                                                                   |
| <a href="#">bn.wikipedia.org</a>          | L3   | 2    | 2       | 26, 32                                                               |
| <a href="#">en.banglapedia.org</a>        | L3   | 6    | 6       | 18, 19, 22, 33, 47, 55                                               |
| <a href="#">en.m.wikipedia.org</a>        | L3   | 1    | 1       | 40                                                                   |
| <a href="#">en.wikipedia.org</a>          | L3   | 18   | 18      | 1, 2, 10, 12, 15, 20, 21, 25, 27, 29, 36, 38, 41, 51, 52, 56, 57, 61 |
| <a href="#">greenbelt.com.bd</a>          | L3   | 1    | 1       | 49                                                                   |
| <a href="#">panchagarh.info</a>           | L3   | 1    | 1       | 44                                                                   |
| <a href="#">prothomalo.com</a>            | L3   | 1    | 1       | 56                                                                   |
| <a href="#">scribd.com</a>                | L3   | 1    | 1       | 61                                                                   |
| <a href="#">thedailystar.net</a>          | L3   | 2    | 2       | 3, 10                                                                |
| <a href="#">touristplaces.com.bd</a>      | L3   | 1    | 1       | 26                                                                   |
| <a href="#">trip.com</a>                  | L3   | 1    | 1       | 4                                                                    |
| <a href="#">ukessays.com</a>              | L3   | 1    | 1       | 29                                                                   |
| <a href="#">vromonguide.com</a>           | L3   | 1    | 1       | 31                                                                   |
| <a href="#">commons.wikimedia.org</a>     | L4   | 2    | 2       | 8, 30                                                                |
| <a href="#">dreamstime.com</a>            | L4   | 1    | 1       | 61                                                                   |

The table shows the actual provenance mix in the loaded museum-source relation. Level 3 sources are retained because they document the research and discovery process, but they are not described as equivalent to official museum or government evidence. A museum having a Level 3 source in the current source relation therefore becomes an explicit review target for future primary-source strengthening.

At museum level, 13 of the 63 museums have at least one Level 1 source row, one of those museums also has a Level 2 supporting source, 29 museums are represented only by Level 3/Level 4 source rows in the current `source.csv`, and 21 museums have no row in the source relation. This is a **source-coverage finding**, not a claim that those 21 museums have no information elsewhere in the project; for example, the separate `contact.csv` contains website values for 29 museums. The distinction is important because the relational `source` table is the explicit provenance relation used for this audit.

### 3.3.2 Artifact-Level Source Audit

Artifact provenance is materially stronger for the Liberation War Museum records. The `v6 artifact.csv` contains 3,505 artifact records, of which 2,406 have a nonblank `Source_Link`. Among those linked artifact rows, 2,396 (99.6%) point to the Liberation War Museum domain, while 10 (0.4%) use secondary domains. The ten secondary-linked records are retained rather than silently deleted; they are explicitly identifiable review targets. The audit therefore avoids the unsupported statement that every artifact is backed by a primary source while demonstrating that the large LWM artifact block has first-party provenance.

Table 12: Artifact-level provenance audit for nonblank `Source_Link` values

| Domain / class                                    | Tier | Artifact rows | Interpretation                                                                                        |
|---------------------------------------------------|------|---------------|-------------------------------------------------------------------------------------------------------|
| Liberation War Museum domain                      | L1   | 2,396         | First-party museum provenance for the linked artifact records.                                        |
| Wikipedia / Banglapedia / tourism / history sites | L3   | 10            | Secondary supporting provenance; retain but prioritize replacement/cross-check with primary evidence. |
| <b>Total nonblank artifact Source_Link</b>        |      | <b>2,406</b>  | 68.6% of the 3,505 artifact records.                                                                  |

The audit shows that third-party links are present in the artifact data. They are listed separately from first-party museum links so that the source type is clear.

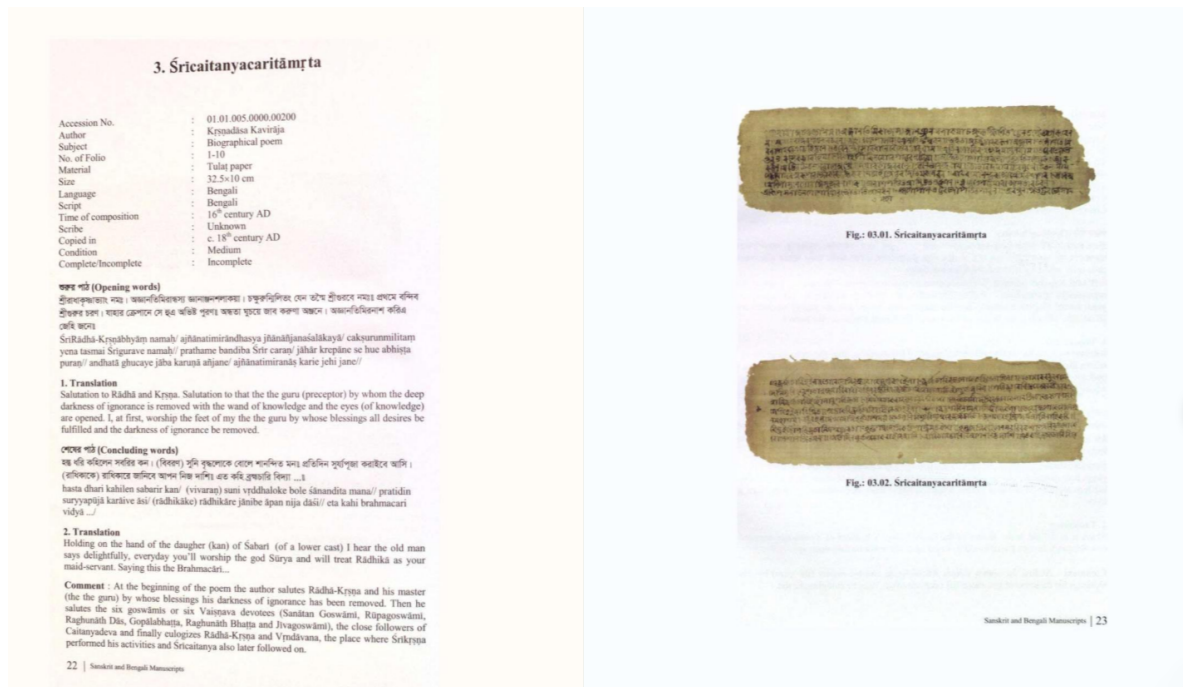


Figure 8: Example page from the Bangladesh National Museum manuscript catalogue represented in the artifact list as one catalogue record (Artifact ID 9).

This page is an example from the catalogue recorded in our artifact list as a single item: A *descriptive catalogue of the Sanskrit and Bengali manuscripts* (Artifact ID 9). The page shown here contains one manuscript entry with catalogue details and photographs. We did not add the individual manuscripts shown on this page as separate artifact records.

**Catalogue source:** <https://archive.org/details/sanskrit-and-bengali-manuscripts-in-bangladesh-national-museum-pt-1/Sanskrit%20and%20Bengali%20manuscripts%20in%20Bangladesh%20National%20Museum%20Pt-1/page/22/mode/2up>

### 3.3.3 Source-Validation Limitations and Action

The audit does not fabricate a “fully primary” dataset. It records the current state and makes the remaining source-quality work visible. The recommended maintenance rule is: **use Level 1 or Level 2 evidence for factual database values whenever available; retain Level 3 sources for discovery/context; use Level 4 primarily for visual support; and mark unresolved provenance for review rather than inventing a value.** This policy is consistent with the project’s missing-value policy: if a value cannot be verified, it remains missing rather than being estimated.

### 3.3.4 Museums Discovered

The museum-level dataset contains 63 documented museums across all eight divisions of Bangladesh. The complete list below is used as the project’s museum master list:

1. Ahsan Manzil (Old Dhaka)
2. Bagha Museum (Bagha)

3. Bangabandhu Boat Museum (Patharghata)
4. Bangladesh Air Force Museum (Tejgaon)
5. Bangladesh Maritime Museum (Chittagong)
6. Bangladesh Bank Taka Museum (Mirpur-2)
7. Bangladesh Military Museum (Bijoy Sarani (Tejgaon))
8. Bangladesh National Museum (Shahbag)
9. Barisal Divisional Museum (Barisal District)
10. Birshreshtha Munshi Abdur Rouf Library and Museum (Roufnagar)
11. Mohammad Ali Palace Museum (Bogra)
12. Bangabandhu Memorial Museum (Dhanmondi)
13. Birisiri Upajatiya Cultural Academy Museum (Birisiri (Durgapur))
14. Chandpur Fisheries Museum (Chandpur)
15. Comilla Rammala Museum (Cumilla)
16. Cox's Bazar Fisheries Museum (Cox's Bazar)
17. Children's Museum (Shishu Academy) (Old High Court Area)
18. Dhaka Nagar Jadughar (Nagar Bhaban (5th Floor))
19. Dinajpur Museum (Dinajpur)
20. Ethnological Museum of Chittagong (Agrabad C/A)
21. Fish Museum & Biodiversity Centre (Mymensingh)
22. Folk Heritage Museum (Bangla Academy) (Shahbagh)
23. Faridpur Museum (Faridpur)
24. Jamalpur Estate Museum (Thakurgaon)
25. Kangal Harinath Memorial Museum (Kumarkhali)
26. Kushtia Municipal Museum (Kushtia)
27. Khulna Divisional Museum (Khulna)
28. Kishoreganj Museum (Kishoreganj)
29. Lalbagh Fort (Lalbagh)
30. Liberation War Museum (Agargaon)
31. Lokayon Museum (Moulvibazar)
32. Madaripur Museum (Madaripur)
33. Archaeological Museum of Mahasthangarh (Shibganj)
34. Mainamati Museum (Comilla)
35. Michel Modhusudhan Dutt Museum (Sagardari)

36. Museum of Rajas' (Zinda Bazar)
37. Museum of Independence, Dhaka (Suhrawardy Udyan)
38. National Museum of Science and Technology (Dhaka)
39. Museum of Geological Survey of Bangladesh (Dhaka)
40. Mymensingh Museum (Mymensingh)
41. National Art Gallery (Bangladesh) (Shahbag)
42. Natore Rajbari (Natore)
43. Osmani Museum (Sylhet)
44. Panchagarh Rocks Museum (Panchagarh)
45. Philatelic Museum (Dhaka)
46. Bangladesh Police Liberation War Museum (Rajarbagh)
47. Postal Museum (Dhaka)
48. Rangpur Museum (Rangpur)
49. Sunamganj Hasan Raja Museum (Sunamganj)
50. Sonargaon Bangladesh Folk Art Museum (Sonargaon)
51. Shilaidaha Kuthibari Memorial Museum (Shilaidaha)
52. Shahjadpur Kacharibari Memorial Museum (Shahzadpur)
53. Tajhat Palace (Rangpur)
54. Tribal Museum (Rangamati)
55. Varendra Research Museum (Rajshahi)
56. Zainul Abedin Museum (Mymensingh)
57. Zia Memorial Museum (Chittagong)
58. Sreemangal Tea Resort and Museum (Sreemangal)
59. Bengal Center (Dhaka)
60. K.B. Ahsaah (R:) Museum (Old Dhaka)
61. Paharpur Archaeological Museum (Badalgachhi)
62. Chittagong University Museum (Chittagong)
63. Baliati Palace Museum (Saturia)

### **3.3.5 Artifact Categories Discovered**

The artifact research reveals several recurring categories. These categories are used during data organization and later inform the Artifact Type relation.

1. Sculptures – Hindu, Buddhist, and Jain stone sculptures from different historical periods.
2. Terracotta – architectural plaques, figurines, and decorative elements.

3. Bronze images – Buddhist and Hindu bronze sculptures.
4. Coins – ancient, medieval, and modern coins, including punch-marked, Gupta, Sultanate, and Mughal issues.
5. Manuscripts – palm-leaf and paper manuscripts in Sanskrit, Bengali, Arabic, and Persian.
6. Weapons – swords, shields, cannons, and firearms.
7. Paintings – miniature paintings and modern art.
8. Textiles – muslin, Nakshi Kantha, and traditional garments.
9. Jewelry – silver, gold, and bronze ornaments.
10. Natural history specimens – taxidermied animals, fossils, and botanical specimens.
11. Historical documents – copperplate inscriptions, land grants, and personal letters.
12. Furniture – zamindari furniture and palace furnishings.

### **3.4 Challenges in Sprint 1**

Data acquisition proceeds museum by museum. For each museum, we record location and administrative details such as district, division, category, ownership, opening hours, entry fee, and contact information. Artifact information is collected after museum-level information is established, because artifact records must remain associated with the correct museum and, where applicable, the correct gallery.

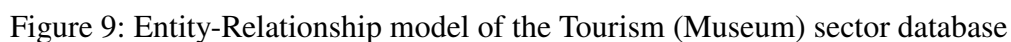
The first major challenge is source fragmentation. A comprehensive list of museums can be constructed, but detailed information about an individual museum is often distributed across several sources. The second challenge is inconsistent formatting. Opening hours, fees, ownership descriptions, and other fields are presented differently by different sources and therefore require manual reconciliation.

The third challenge is artifact image collection. Reliable images for individual artifacts are not consistently published online, making image collection one of the most time-consuming tasks. The fourth challenge is incomplete attribute values. Period dating, finding place, dimensions, and image links may be unavailable or insufficiently precise. We therefore record verified information and keep unsupported values missing instead of inventing them.

A further challenge is conflicting information. When two sources disagree, we cross-check the records and evaluate the reliability of the sources before accepting a value. Finally, artifact-level documentation does not scale evenly across all museums. Some museums provide enough public information to support detailed artifact tables, while others provide only museum-level information.

By the end of the data-collection and normalization stage, the working dataset contains 63 museum records and 3,505 artifact records, with the source analysis covering 89 identified sources. The collected information is organized into museum-level and artifact-level structures and then transformed into normalized CSV relations for MySQL implementation.

The Entity-Relationship model is derived from the structure of the collected data. We identify the information that must be represented independently and then determines how those information components are related. Figure 9 presents the resulting conceptual model.



The Museum entity is central to the model. It connects to Owner through owned\_by and to Category through classified\_as. A Museum consists\_of Gallery entities, while galleries exhibit artifacts. Museum records also connect to Contact, Entry Fee, Opening Hours, Closed Days, and Source so that visiting information, communication details, and source provenance remain structured rather than being mixed into one large record.

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and artifact\_has\_material. The Image relation supports multiple image records for an artifact, and Source preserves provenance links used during research.

### 3.6 Entities

The main entity types identified from the collected information are summarized in Table 13. The table is a compact description of the conceptual model, while Figure 9 provides the full visual relationship structure.

Table 13: Main entities and their purposes

| Entity        | Purpose                                                                                  |
|---------------|------------------------------------------------------------------------------------------|
| Division      | Represents the highest geographic level used for museum location.                        |
| District      | Represents a district within a division.                                                 |
| City          | Represents a city associated with a district and used to locate museums.                 |
| Museum        | Represents a museum and its main administrative and location-related attributes.         |
| Gallery       | Represents a gallery belonging to a museum and provides a place for exhibited artifacts. |
| Artifact      | Represents an individual museum artifact and its descriptive characteristics.            |
| Donor         | Represents a donor or credit associated with one or more artifacts.                      |
| Owner         | Represents the ownership information associated with a museum.                           |
| Category      | Represents the category or classification of a museum.                                   |
| Artifact Type | Represents the type or classification of an artifact.                                    |
| Material      | Represents the material from which an artifact is made.                                  |
| Image         | Represents an image associated with an artifact and its source link.                     |
| Entry Fee     | Represents museum fee types and fee amounts.                                             |
| Opening Hours | Represents opening and closing times for specific days.                                  |
| Closed Days   | Represents the days on which a museum remains closed.                                    |
| Contact       | Represents website, phone, and email information for a museum.                           |
| Source        | Represents the source URL used to support museum information.                            |

Table 13 shows how the conceptual model separates geographic, museum, artifact, administrative, visiting, and provenance concerns. The separation is important because these information groups have different structures and update patterns.

### 3.7 Relationships

The principal relationships represented in Figure 9 are summarized in Table 14. Each relationship connects two or more entities according to the meaning observed in the collected data.

Table 15 makes the key-based implementation of the revised Entity-Relationship model ex-

Table 14: Principal relationships in the Entity-Relationship model

| Relationship      | Connected entities     | Meaning                                                                                                                                                        |
|-------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| includes          | Division–District      | A division includes districts.                                                                                                                                 |
| comprises         | District–City          | A district comprises cities.                                                                                                                                   |
| hosts             | City–Museum            | A city hosts museums.                                                                                                                                          |
| consists_of       | Museum–Gallery         | A museum consists of galleries.                                                                                                                                |
| exhibits          | Gallery–Artifact       | A gallery exhibits artifacts associated with the museum.                                                                                                       |
| made_of           | Artifact–Material      | An artifact may be associated with one or more materials; the M:N relationship is normalized through artifact_has_material.                                    |
| belongs_to        | Artifact–Artifact Type | An artifact may have one or more artifact types; the M:N relationship is normalized through artifact_has_type.                                                 |
| credited_to       | Artifact–Donor         | An artifact may have one or more donors, and a donor may be associated with multiple artifacts; the M:N relationship is normalized through artifact_has_donor. |
| represented_by    | Artifact–Image         | An artifact can have multiple available images.                                                                                                                |
| owned_by          | Museum–Owner           | A museum is associated with an owner.                                                                                                                          |
| classified_as     | Museum–Category        | A museum is classified into a category.                                                                                                                        |
| charges           | Museum–Entry Fee       | A museum has entry fee records.                                                                                                                                |
| operates_during   | Museum–Opening Hours   | A museum has opening hours by day.                                                                                                                             |
| remains_closed_on | Museum–Closed Days     | A museum has specified closed days.                                                                                                                            |
| provides          | Museum–Contact         | A museum provides contact information.                                                                                                                         |
| references        | Museum–Source          | Museum information is linked to source references.                                                                                                             |

Table 15: Foreign-key mapping derived from the Entity-Relationship model

| Relation              | Foreign Key                   | References                     |
|-----------------------|-------------------------------|--------------------------------|
| District              | Division_ID                   | Division(Division_ID)          |
| City                  | District_ID                   | District(District_ID)          |
| Museum                | City_ID                       | City(City_ID)                  |
| Museum                | Owner_ID                      | Owner(Owner_ID)                |
| Museum                | Category_ID                   | Category(Category_ID)          |
| Gallery               | Museum_ID                     | Museum(Museum_ID)              |
| Artifact              | (Museum_ID, Gallery_No)       | Gallery(Museum_ID, Gallery_No) |
| Artifact              | Dimension_ID                  | Dimension(Dimension_ID)        |
| Artifact Has Donor    | Artifact_ID, Donor_ID         | Artifact, Donor                |
| Artifact Has Type     | Artifact_ID, Artifact_Type_ID | Artifact, Artifact Type        |
| Artifact Has Material | Artifact_ID, Material_ID      | Artifact, Material             |
| Image                 | Artifact_ID                   | Artifact(Artifact_ID)          |
| Museum Phone          | Museum_ID                     | Museum(Museum_ID)              |
| Entry Fee             | Museum_ID                     | Museum(Museum_ID)              |
| Source                | Museum_ID                     | Museum(Museum_ID)              |
| Opening Hours         | Museum_ID                     | Museum(Museum_ID)              |
| Closed Days           | Museum_ID                     | Museum(Museum_ID)              |
| Contact               | Museum_ID                     | Museum(Museum_ID)              |

plicit. The M:N Artifact–Donor, Artifact–Artifact Type, and Artifact–Material relationships are implemented through junction tables, while Gallery retains its composite museum-specific key.

The relationships in Table 14 show that the database is not a collection of independent tables. The geographic hierarchy supports location-based queries, the museum-gallery-artifact structure supports collection queries, and the administrative and provenance relations preserve details needed for tourism use and source verification.

### 3.8 Data Population

The prepared CSV files act as the source data for the relational implementation. The current dataset contains 63 museum records and 3,505 artifact records, while the source analysis covers 89 identified sources. The normalized CSV package was loaded into MySQL Workbench after pre-flight validation, and the populated database was subsequently checked using SQL record-count, relationship, and referential-integrity queries.

### 3.9 ETL Pipeline Development

The collected CSV data were extracted, cleaned, transformed, validated, and loaded into MySQL. The implemented ETL workflow is designed to preserve the collected information while making it suitable for the normalized relational schema. The process consists of five connected stages: Extract, Transform and Normalize, Pre-flight Validation, Load, and Post-load Validation.

1. **Extract:** The Python loader reads the prepared CSV files and maps each file to its corresponding relational table. The extracted records include museum, gallery, artifact, donor, artifact type, material, image, phone, fee, opening-hour, closed-day, contact, source, and geographic information.
2. **Transform and Normalize:** Values are cleaned and standardized for insertion. Duplicate records and invalid references are checked. Missing values are retained as NULL when the source does not provide a verified value. Multi-valued artifact relationships are separated into junction tables so that repeating groups are not stored inside one artifact row.
3. **Pre-flight Validation:** Before insertion, required fields and selected foreign-key references are checked. The loader stops if the prepared dataset fails these checks. The completed run produced the message: “Preflight OK: all required CSV values and checked FK references are valid.”
4. **Load:** The validated records are inserted into MySQL in dependency order. Parent geographic and museum records are loaded before dependent artifact and relationship records. Python uses the MySQL connector to establish the database connection and perform the insert operations.
5. **Post-load Validation:** SQL queries are used to verify record counts, foreign-key integrity, normalized many-to-many relationships, and research-oriented analytical results.

Table 16: Implemented ETL stages and validation activities

| Stage                 | Activity                                                                                                                     | Evidence / Output                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Extract               | Read prepared CSV files and detect their columns and records.                                                                | Structured input records for transformation. |
| Transform             | Clean values, check duplicates, validate keys, retain verified NULLs, and standardize schema-compatible fields.              | Cleaned and schema-compatible records.       |
| Normalize             | Separate multi-valued donor, type, and material relationships into junction tables; store multiple museum phones separately. | Normalized relational structure.             |
| Pre-flight Validation | Check required values and selected foreign-key references before insertion.                                                  | Preflight OK.                                |
| Load                  | Insert parent, child, and junction tables into MySQL using Python and mysql-connector.                                       | Populated MySQL database.                    |
| Post-load Validation  | Run SQL count, relationship, orphan, and analytical queries.                                                                 | Validation results and research evidence.    |

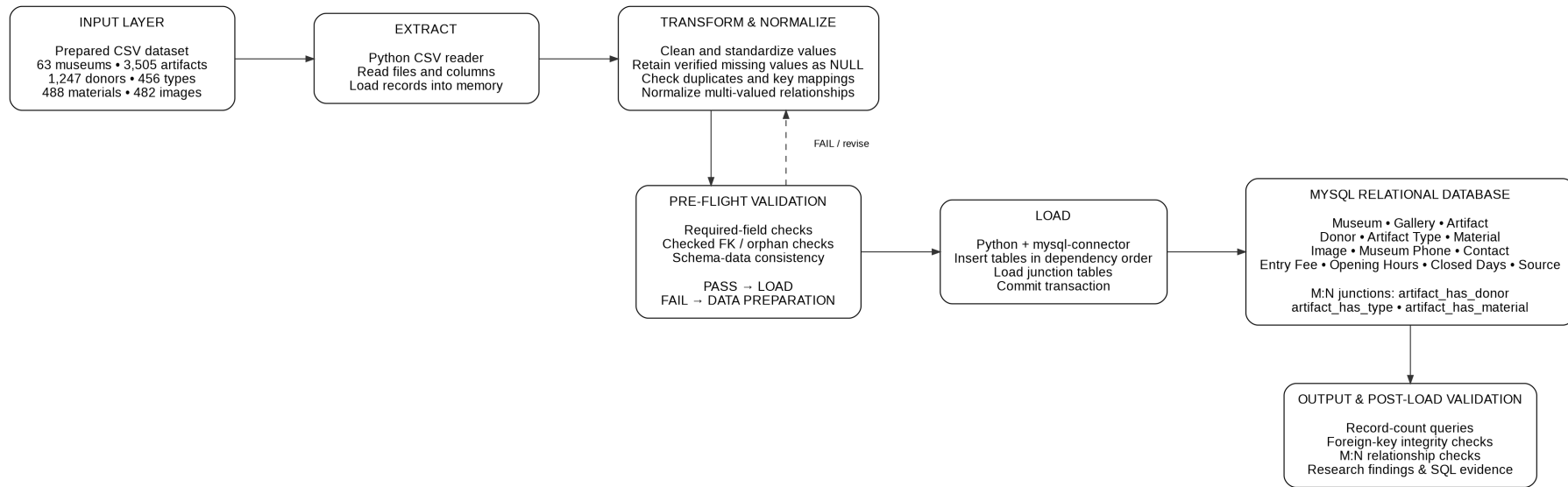


Figure 10: Implemented ETL pipeline architecture for the Bangladesh Tourism (Museum) relational database

Figure 10 summarizes the implemented workflow from prepared CSV input to extraction, transformation and normalization, pre-flight validation, MySQL loading, and post-load SQL validation. A failed validation is returned to data preparation rather than being silently inserted into the database.

The successful MySQL load produced the following principal counts: 63 museums, 338 galleries, 3,505 artifacts, 1,247 donors, 456 artifact types, 488 materials, 3,127 artifact-donor links, 4,094 artifact-type links, 1,877 artifact-material links, 482 images, 15 museum-phone records, 53 entry-fee records, 21 opening-hour records, 20 closed-day records, 29 contact records, and 74 source records.

```

1 • USE bangladesh_tourism_museum;
2
3 • SELECT 'Museum' AS Table_Name, COUNT(*) AS Record_Count FROM museum
4 UNION ALL
5 SELECT 'Gallery', COUNT(*) FROM gallery
6 UNION ALL
7 SELECT 'Artifact', COUNT(*) FROM artifact
8 UNION ALL
9 SELECT 'Donor', COUNT(*) FROM donor
10 UNION ALL
11 SELECT 'Artifact Type', COUNT(*) FROM artifact_type
12 UNION ALL

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

|   | Table_Name    | Record_Count |
|---|---------------|--------------|
| ▶ | Museum        | 63           |
|   | Gallery       | 338          |
|   | Artifact      | 3505         |
|   | Donor         | 1247         |
|   | Artifact Type | 456          |
|   | Material      | 488          |
|   | Image         | 482          |
|   | Museum Phone  | 15           |
|   | Source        | 74           |

Figure 11: MySQL Workbench record-count validation after ETL loading

01\_create\_tables 01\_create\_tables\* 01\_create\_tables\* 01\_create\_tables 01\_create\_tables 01\_create\_tables

the connected server

Limit to 1000 rows

```

1 • SELECT
2     a.Artifact_ID,
3     a.Artifact_Name,
4     COUNT(DISTINCT atp.Artifact_Type_ID) AS Number_of_Types,
5     GROUP_CONCAT(
6         DISTINCT t.Artifact_Type_Name
7         ORDER BY t.Artifact_Type_Name
8         SEPARATOR ', '
9     ) AS Artifact_Types
10  FROM artifact a
11  JOIN artifact_has_type atp
12      ON a.Artifact_ID = atp.Artifact_ID
13  JOIN artifact_type t
14      ON atp.Artifact_Type_ID = t.Artifact_Type_ID
15  GROUP BY a.Artifact_ID, a.Artifact_Name

```

Result Grid

|   | Artifact_ID | Artifact_Name                                      | Number_of_Types | Artifact_Types                              |
|---|-------------|----------------------------------------------------|-----------------|---------------------------------------------|
| ▶ | 16          | Mahamaya                                           | 3               | Hindu Sculpture, Shaiva, Shakta             |
|   | 156         | Traditional Garments: Choka and Chapkan            | 3               | Decorative Arts, Textile, Traditional Dress |
|   | 162         | Parnashabari — Tribal Goddess Wearing Leaves       | 3               | Buddhist Sculpture, Tantric, Vajrayana      |
|   | 163         | Pratisara / Mahapratishara — The Great Protectr... | 3               | Buddhist Sculpture, Pancaraksha, Vajrayana  |
|   | 165         | Namasangiti — Female Form (Namasangiti-devi)       | 3               | Buddhist Sculpture, Tantric, Vajrayana      |
|   | 168         | Gauri with Infant Sadyojata Shiva                  | 3               | Hindu Sculpture, Shaiva, Shakta             |
|   | 187         | Pakistani Rupee — Coins & Notes (1947–1971)        | 3               | Banknote, Coin, Historical Currency         |
|   | 197         | World Currency Collection — ~200 Countries (G...   | 3               | Banknote, Coin Collection, World Currencies |
|   | 213         | Hajong Traditional Costumes & Jewelry              | 3               | Ethnographic Artifact, Jewelry, Textile     |
|   | 226         | Comparative Religious Texts — Multi-Faith Colle... | 3               | Book, Manuscript, Religious Studies         |

Result 2 x

Output

Action Output

# Time Action Message

Figure 12: SQL evidence of artifacts associated with multiple artifact types

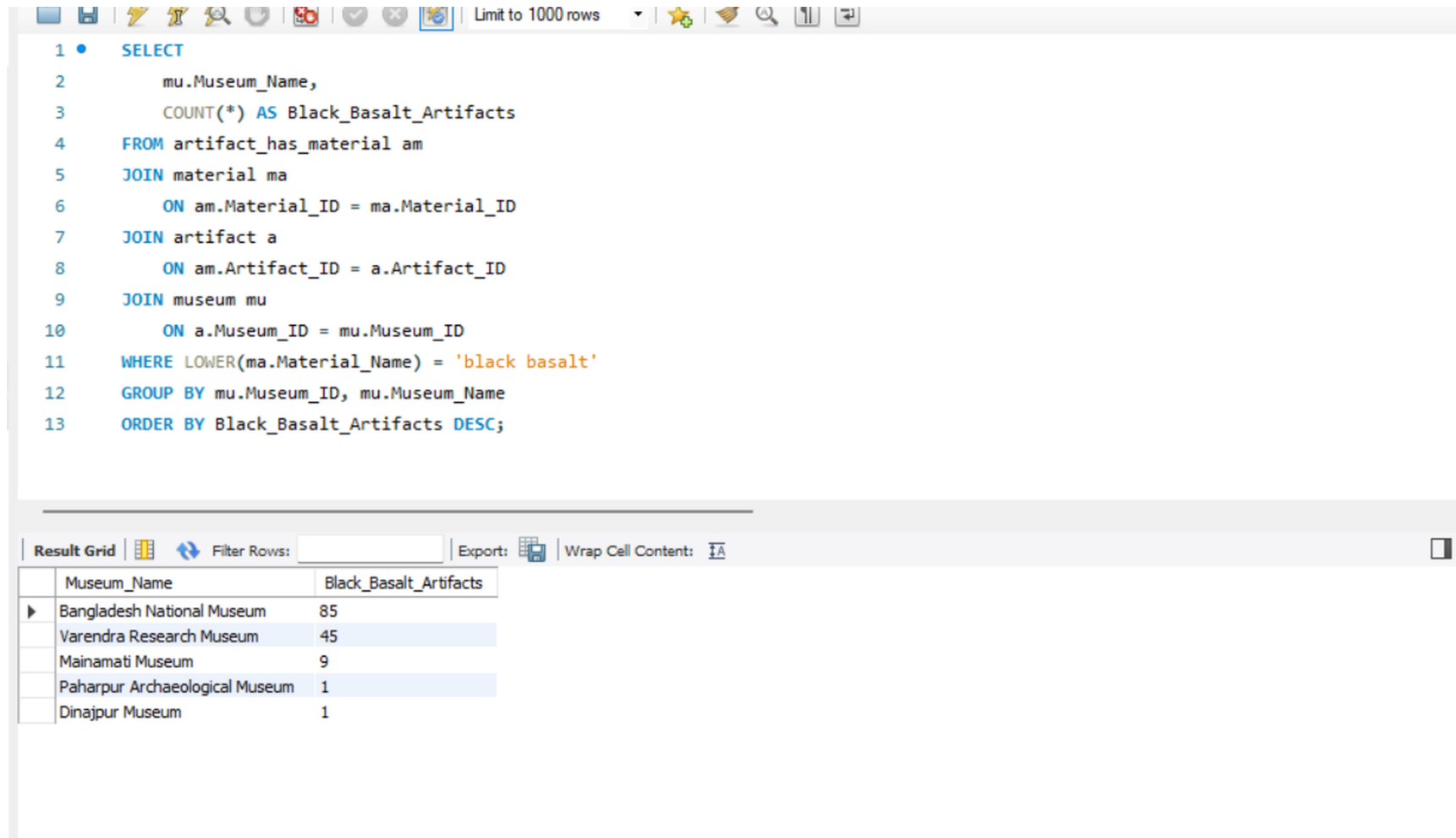


Figure 13: SQL evidence of Black Basalt artifact distribution by museum

Figures 11–13 provide implementation evidence captured from MySQL Workbench. The record-count result confirms successful population of the principal tables, the artifact-type query demonstrates the practical need for a normalized M:N relationship, and the material query demonstrates that the normalized database can support collection-level research findings.

Table 17: ETL issues identified during implementation and their resolutions

| Issue observed                                                | Why it mattered                                                                               | Resolution in the implementation                                                                           |
|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Mixed gallery-count values such as numeric and narrative text | A numeric SQL type could not represent all source values without truncation or failure.       | The gallery-count field was made text-compatible so source values could be retained.                       |
| Mixed-format entry-fee descriptions                           | Fee information can contain narrative text rather than a single numeric amount.               | The fee amount field was stored as text-compatible data instead of forcing unsupported numeric conversion. |
| Blank artifact rows detected in the prepared CSV              | Required artifact names cannot be inserted into a NOT NULL <code>Artifact_Name</code> column. | The empty artifact rows were removed from the prepared load set rather than inventing artifact names.      |
| Missing gallery references detected during preparation        | An Artifact foreign key to Gallery must resolve to an existing museum-specific gallery key.   | The missing referenced gallery records were added using the collected gallery labels before loading.       |
| Source column-name mismatch during loader validation          | The CSV and SQL schema used different names for the source-link field.                        | The implementation was aligned to the actual CSV field name, <code>Source_Link</code> .                    |
| Explicit source placeholder such as none                      | Treating a literal source value as NULL would silently change the collected information.      | The loader preserves explicit source-provided placeholders rather than converting them to NULL.            |

The purpose of documenting these issues is to show that ETL was an iterative validation process rather than a single blind import. Each problem was identified from an actual load or pre-flight check, and the resolution was chosen to preserve the collected information without fabricating unsupported facts.

### 3.9.1 Validation Evidence and Results

The teacher feedback identified validation as a missing piece of the report. The final implementation therefore separates validation into structural, referential, provenance, and relationship checks. The results below are generated from the same v6 CSV package used by the ETL loader and are also represented as reproducible SQL in `04_validation_and_source_audit.sql`.

Table 18: Pre-load and post-load validation audit of the v6 dataset

| Validation check                      | Result                    | Interpretation                                                                                                                                    |
|---------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Required fields                       | 0 blank required values   | Required identifiers and mandatory names/relationship keys are populated in the prepared CSVs.                                                    |
| Exact duplicate rows                  | 0 duplicate rows          | No exact duplicate CSV records were found in the audited relations.                                                                               |
| Checked foreign-key references        | 0 orphan references       | All checked parent-child references resolve to an existing parent record.                                                                         |
| Artifact–Gallery composite references | 0 unresolved              | Every nonblank artifact gallery reference matches (Museum_ID, Gallery_No) in gallery.                                                             |
| Source URL syntax                     | 0 malformed in source.csv | All 74 museum-source links have an HTTP(S) URL form.                                                                                              |
| Artifact Source_Link coverage         | 2,406 / 3,505 (68.6%)     | A substantial subset has explicit artifact-level provenance; remaining records are not assigned an invented source.                               |
| Image Source_Link coverage            | 458 / 482                 | Most image records contain a source link; 5 source-link values need manual formatting review.                                                     |
| Image references                      | 49 non-HTTP references    | These are filename/path-like image references and should be reviewed before external deployment; they are not silently treated as valid web URLs. |

The source audit also identifies a limitation that should be visible rather than hidden: five image source-link values are malformed/non-URL strings in the current CSV, and 49 image references are filename/path-like rather than HTTP(S) URLs. These values are not changed by the audit because doing so would require inventing or guessing the intended links. They are flagged as maintenance items.

Table 19: Actual M:N and multi-valued evidence supporting normalization

| Observed relationship                  | Count | Normalized relation   |
|----------------------------------------|-------|-----------------------|
| Artifacts with multiple donors         | 111   | artifact_has_donor    |
| Artifacts with multiple artifact types | 571   | artifact_has_type     |
| Artifacts with multiple materials      | 427   | artifact_has_material |
| Artifacts with multiple images         | 8     | image                 |
| Museums with multiple phone records    | 3     | museum_phone          |

These checks provide a concrete bridge between normalization and validation. The junction tables are not merely theoretical constructs: the loaded data contain hundreds of genuine multi-

valued artifact relationships, and SQL grouping confirms that those relationships are represented as separate rows. The validation SQL also checks that each junction-row identifier resolves to an existing artifact and master record.

### **3.9.2 AI-Assisted Work and the Data Authenticity Boundary**

We used AI tools during source discovery, information extraction, organization, and repetitive checking. They helped us find possible pages, locate information in long sources, organize fields, and notice possible inconsistencies. We did not use an AI response as the source for a museum or artifact value. Before keeping a value, we checked it against the source we recorded. When a value could not be supported, we left it blank rather than filling it from an AI suggestion.

For this reason, we kept source checking and database validation as separate checks. A record can have a URL even when the source is secondary, and a row can pass primary-key and foreign-key checks while some optional fields are still blank.

single word “verified” for all of them.

### **3.10 Data Quality and Evidence Boundary**

The implementation evidence in this report is limited to artifacts and execution results that can be supported by the project data and captured database outputs. The ETL section documents the completed loading workflow, while the SQL evidence figures show actual MySQL Workbench results. Source-provided missing values are not converted into invented facts, and explicit source placeholders are preserved by the loader when they are part of the collected data.

## **4 Deployment and Maintenance**

This chapter describes the implemented deployment and maintenance approach. Data collection and database modelling are followed by MySQL Workbench schema creation, ETL database population, SQL query development, and validation.

### **4.1 Deployment Strategy**

The relational schema has been implemented in MySQL Workbench. Deployment begins with database creation and the definition of the relational tables, primary keys, foreign keys, and integrity constraints. The validated CSV data is then loaded in dependency order so that parent records exist before dependent and junction records.

After population, the database was checked through representative Structured Query Language

operations involving museum, artifact, geographic, visiting, contact, fee, and source information. The implemented query workload includes basic retrieval, filtering, joins, and aggregation. Example questions include finding museums by district or city, retrieving artifacts by material or period, comparing entry fees, checking opening information, counting museums by geographic area, and identifying museums containing artifacts of a particular type.

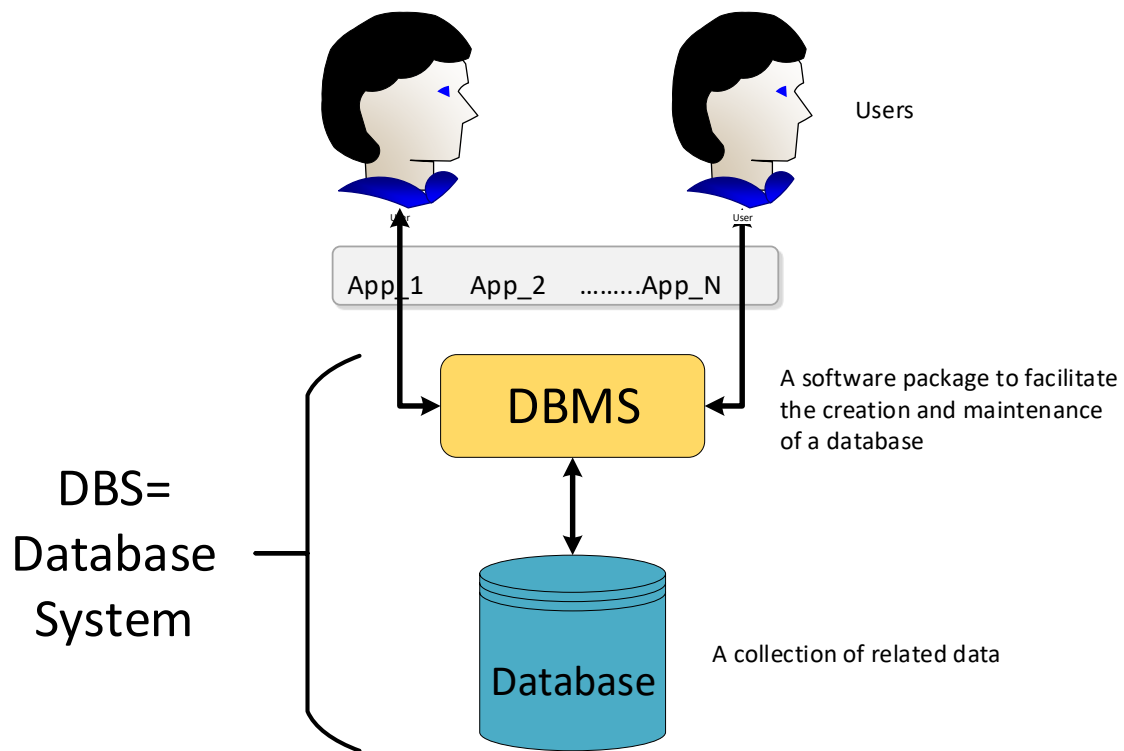


Figure 14: General Database System architecture for the implemented database environment

Figure 14 illustrates the general Database System architecture used by the project. Users interact with applications or interfaces rather than directly manipulating the stored data. The application layer communicates with the Database Management System, which manages the database and executes Structured Query Language operations. In the current project, MySQL Workbench is the implemented environment for database creation, loading, and query testing; a user-facing application remains future work.

## 4.2 Maintenance Procedures

The museum dataset requires continuing maintenance because public information can change and new sources can provide additional artifact details. We preserve source references so that updates can be traced back to evidence. New information can be reviewed, incorporated into the shared dataset, and then loaded into the database without changing the conceptual structure unless a new requirement justifies a model revision.

Maintenance also includes periodic review of missing artifact images, dimensions, period information, finding places, and other fields that remain unavailable at the current stage. The

project treats a verified missing value as preferable to an unsupported estimate. When a new reliable source becomes available, the relevant record can be updated and revalidated before database loading.

### 4.3 Code and Data Repository

The shared Google Drive and Google Sheets environment acts as the current working repository for the dataset, collected images, source information, and museum-specific research tables. The documentation source is maintained collaboratively in Overleaf. The implementation package contains SQL schema scripts, normalized CSV data, and Python data-loading resources so that the database can be reproduced and updated systematically.

Listing 1: Actual GitHub repository structure

```
Bangladesh–Museum–Tourism–Database /
|-- README.md
|
|-- database /
|   |-- schema /
|   |   '-- 01_create_tables.sql
|   |
|   |-- data /
|   |   '-- CSV data files
|   |
|   '-- queries /
|       |-- 01_validation.sql
|       |-- 02_basic_queries.sql
|       '-- 03_finding_queries.sql
|
|-- etl /
|   |-- load_database.py
|   |-- requirements.txt
|   '-- README.md
|
|-- documentation /
|   |-- ER_Diagram.png
|   |-- ER_Diagram.pdf
|   |-- ETL_Documentation.pdf
|   '-- Finding_Table.pdf
|
|-- evidence /
|   |-- preflight_success.png
|   |-- mysql_load_success.png
|   '-- sql_validation.png
```

Listing 1 records the repository structure shown for the project. The organization separates

the database schema, CSV data, SQL query scripts, ETL loader, documentation, and execution evidence. This separation supports reproducibility: the schema can be recreated independently, the prepared CSV data can be reloaded through the ETL script, validation and finding queries are kept separately from the loader, and screenshots provide visible evidence of execution.

The repository also contains a project-level README.md and an ETL-specific README.md. The project README acts as the entry point for the repository, while the ETL README provides implementation-oriented guidance for the loading component. The repository therefore documents not only the final database output but also the resources needed to understand and reproduce the implementation.

#### 4.3.1 Some Important Findings from the Collected Data

The following findings are derived by comparing artifact type, period, material, finding place, provenance, and museum records in the collected dataset. The purpose is to highlight historical and cultural patterns that emerge from the data.

Table 20: Some Important Findings from the Museum and Artifact Dataset

| No. | Finding                                                                                                                                                                                                                                        | Evidence (counts, from the actual data)                                                                                                                                                                                                                                                                                                        |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | 93%→91% of all sculpture-type artifacts nationwide date to the Pala–Sena period (7th–13th c. CE)                                                                                                                                               | Across all sculpture-related types (Sculpture, Hindu/Buddhist/Bronze/Stone/Stucco/Metal/Portrait/Architectural Sculpture) = 235 artifacts; 215 of them (91%) fall in the 7th–13th century window. Concentrated at Bangladesh National Museum (104), Varendra Research Museum (46), Paharpur Archaeological Museum (30), Mainamati Museum (29). |
| 2   | Those same sculptures are almost all Black Basalt, and mostly excavated at Paharpur (Naogaon) and Mainamati (Comilla)                                                                                                                          | Black Basalt = single most common material dataset-wide (141 records), 85 of them at Bangladesh National Museum, 45 at Varendra.                                                                                                                                                                                                               |
| 3   | Bangladesh’s numismatic collections form a continuous ~2,400-year timeline – punch-marked coins from Wari-Bateshwar (4th c. BCE) through Mauryan, Gupta, Kushan, Harikela, Sultanate, Mughal, colonial, and the first Bangladeshi paisa (1973) | 36 coin-type artifacts spanning from 4th century BCE to 1973 CE, unbroken.                                                                                                                                                                                                                                                                     |

| No. | Finding                                                                                                                                   | Evidence (counts, from the actual data)                                                                                                                                                                                                                                     |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4   | Most artifacts entered collections through archaeological excavation, not purchase                                                        | Of 616 artifacts with a recorded donor/source, the largest single categories are “Excavation find” (76), “Paharpur Excavation” (19), “K.N. Dikshit Excavation 1930s” (14), Mainamati excavations (8) – excavation-linked entries outnumber “Purchased” (35) by roughly 4:1. |
| 5   | Museums are heavily Dhaka-centric: 26 of 63 museums (41%) sit in Dhaka division, 20 in Dhaka district alone; Barisal division has only 2  | Direct count from museum/city/district/division tables.                                                                                                                                                                                                                     |
| 6   | (Chittagong University Museum) Oldest coin is a Gold Dinar of Umayyad Caliph Abdul Malik, Hijri 77 / AD 696, found in Hajiganj, Chandpur  | Chittagong University Museum record.                                                                                                                                                                                                                                        |
| 7   | (Chittagong University Museum) Islamic manuscripts are Persian-heavy: 111 Persian vs. 40 Arabic vs. 8 illuminated Qurans                  | Chittagong University Museum record.                                                                                                                                                                                                                                        |
| 8   | (Chittagong University Museum) July 2024 Uprising artifacts (martyrs’ clothing, protest graffiti) accessioned the same year as the events | Chittagong University Museum record.                                                                                                                                                                                                                                        |
| 9   | Hindu sculpture is exceptionally concentrated in the 10th–12th century CE                                                                 | 12 of 13 Hindu Sculpture records (92%) are dated to periods including the 10th–12th century CE; all 13 fall within the broader 7th–13th century window.                                                                                                                     |
| 10  | Hindu sculpture is overwhelmingly associated with Black Basalt                                                                            | 10 of 13 Hindu Sculpture records (77%) are recorded as Black Basalt or closely related black-basalt stone materials.                                                                                                                                                        |
| 11  | Buddhist sculpture in the dataset is completely concentrated in the 10th–12th century CE                                                  | 8 of 8 Buddhist Sculpture records (100%) include the 10th–12th century in their dating.                                                                                                                                                                                     |
| 12  | Vajrayana sculpture is also completely concentrated in the same medieval period                                                           | 6 of 6 Vajrayana records (100%) include the 10th–12th century CE.                                                                                                                                                                                                           |

| No. | Finding                                                                                               | Evidence (counts, from the actual data)                                                                                                                                                                                   |
|-----|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13  | Paharpur and Mainamati together form a major archaeological source for the surviving sculpture record | Paharpur-related locations appear in 31 sculpture-related records and Mainamati-related locations in 27; together they account for 58 of 187 sculpture-related artifacts (31%) in the normalized artifact analysis.       |
| 14  | Wari-Bateshwar provides one of the earliest monetary horizons represented in the collection           | The dataset records Silver Punch-Marked Coins (4th–2nd century BCE) associated with Wari-Bateshwar, Narsingdi, and a Narsingdi archaeological coin hoard dated 4th century BCE–2nd century CE.                            |
| 15  | The coin collection can be read as a history of Bengal’s changing political powers                    | The records move through Gupta, Kushan, Harikela, Bengal Sultanate, Mughal, British India, Pakistan-period and Bangladesh currency, including the first Bangladesh 1-Taka note (1972) and first Bangladeshi coins (1973). |

### 4.3.2 SQL Table Creation

The implemented SQL table-creation stage maps the relational schema to MySQL tables. Primary keys provide unique row identification, foreign keys preserve inter-table relationships, and composite keys are used where the model requires more than one attribute to identify a record, such as Gallery, Entry Fee, Source, Opening Hours, Closed Days, Museum Phone, and the M:N junction tables.

### 4.3.3 Extract, Transform and Load

The detailed Extract, Transform, Normalize, Validate, and Load procedure is documented in Section 3.9. The deployment stage uses the validated CSV package and the implemented MySQL schema described there.

### 4.3.4 Load and Validation

The final load stage inserts the transformed records into MySQL Workbench in dependency order. Post-load SQL validation checks record populations, selected foreign-key relationships, normalized many-to-many associations, and research-oriented aggregations.

## 5 Collaboration

The project requires collaboration both within Group-11 and across the wider shared database initiative. The museum dataset is assembled through repeated source checking, so collaboration is necessary for correcting records, sharing discoveries, and maintaining a consistent data structure.

### 5.1 Intra-Level Collaboration (Level-00)

Within Group-11, the four members collaborate through a shared workflow rather than maintaining isolated datasets. Trello provides visibility of tasks, Google Sheets keeps the working records synchronized, WhatsApp supports quick communication, and Overleaf supports collaborative documentation. The shared environment is especially important because museum and artifact records often require correction after cross-checking multiple sources.

The work-stage labels in Figure 5 also support collaboration. A member can identify a Research card that still needs source discovery, a Collection card that requires active information gathering, or a Building card where verified information is being assembled into structured tables. This allows members to contribute to the next available task without requiring every task to be assigned manually.

### 5.2 Inter-Level Collaboration (Scrum of Scrums)

The Tourism sector is one component of the wider MODELBD initiative. Inter-group coordination is therefore important for shared reference structures, especially the geographic hierarchy of Division, District, and City. Maintaining compatible naming and identifiers for these common geographic entities helps the Tourism sector remain integrable with other sector databases.

### 5.3 Good and Bad Memories

The project gave us several positive experiences as well as some difficult moments. We summarize both sides because they reflect how the work actually progressed.

#### Good Memories

- A museum could begin as a simple name on the master list and gradually become a structured record containing location, ownership, category, visiting information, galleries, artifacts, images, and source references.
- Finding a reliable artifact source after searching through several different types of websites gave us a clear sense of progress and made the dataset more useful.
- We enjoyed solving ER-model questions together by comparing the real data with the relationships we wanted the database to represent.
- Learning from each member's strengths and becoming more confident in data cleaning, modeling, and documentation made the project more engaging and practical.

## **Challenges and Lessons Learned**

The main collaboration challenge was keeping a shared dataset consistent while multiple members worked on different museums and sources. A change made after cross-checking one source could affect related records, and artifact information was not equally available for every museum. We therefore learned the importance of communicating corrections, preserving source links, and avoiding unsupported assumptions.

Another lesson was that data collection could consume much more time than expected. We learned that a database project is not only about writing SQL statements; reliable data discovery, verification, structuring, and provenance are essential parts of database development. The project also reinforced the value of early design checking because structural problems identified before implementation are easier to correct than problems discovered after loading.

## **Bad Memories**

- Spending a large amount of time searching for complete and trustworthy artifact information.
- Repeatedly checking image availability for individual artifacts, especially when the same artifact appeared under different names.
- Dealing with inconsistent opening hours, fees, and descriptions across different sources.
- Working under tight academic deadlines while maintaining a large shared dataset.

## 6 Comments on Course Pedagogy

The Database Systems Lab pedagogy differs from a conventional individual or small-scale database assignment because the class contributes to a larger shared data-integration initiative. The Tourism (Museum) project therefore provides experience in database design together with source discovery, collaborative data preparation, Scrum-style task management, and coordination with other groups.

Table 21: Opportunities and limitations of the course pedagogy

| Opportunities                                                                                                                           | Limitations                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Working on one shared, large-scale system gives experience closer to real data integration projects than an isolated classroom dataset. | Coordinating shared conventions, especially common geographic reference data, across several independently working groups can be difficult without a dedicated synchronization mechanism. |
| The Scrum-inspired backlog structure provides practice in incremental and trackable delivery rather than a single final submission.     | Real-world data collection, especially artifact-level research, takes considerably longer than expected and can reduce the time available for implementation and testing.                 |
| Early design checking can catch structural issues before implementation and reduce costly rework.                                       | Publicly available artifact-level information is genuinely scarce for many museums, which limits artifact coverage regardless of the effort invested.                                     |
| The shared repository encourages source traceability and collaborative correction of records.                                           | Working with heterogeneous online sources requires substantial manual reconciliation and verification.                                                                                    |
| The project develops an industry-like sense of responsibility because one group's design decisions can affect future integration.       | The large scope of the shared initiative can make it difficult to balance depth of research with implementation deadlines.                                                                |

### 6.1 Opportunities

The shared sector-based structure provides experience with a larger data-integration problem. We learn to think about common reference entities, source provenance, incremental delivery, and model review rather than focusing only on the syntax of database commands. The Scrum-inspired workflow also provides practical experience in breaking a large problem into smaller tasks and tracking progress visibly.

### 6.2 Limitations

The main limitation is the time required for real-world data collection. Artifact-level information often requires searching several sources and verifying individual records, which can consume the time that would otherwise be available for implementation. Coordination across a larger multi-group structure can also require additional communication when common entities or

conventions are involved.

## 6.3 AI-Assisted Work

We use Artificial Intelligence (AI) tools as supporting resources throughout the project. We do not treat AI output as final evidence. Instead, we use it to speed up discovery, organize information, identify possible gaps, and support repetitive data-preparation tasks, while we open the cited sources ourselves and verify the information before adding it to our dataset or using it in the database design.

The approximate distribution of our AI-assisted work is Claude 41 percent, ChatGPT 33 percent, Copilot 20 percent, and other tools 6 percent. Figure 15 presents this reported distribution. These percentages describe our approximate use of AI assistance and are not automatically measured application-usage statistics.

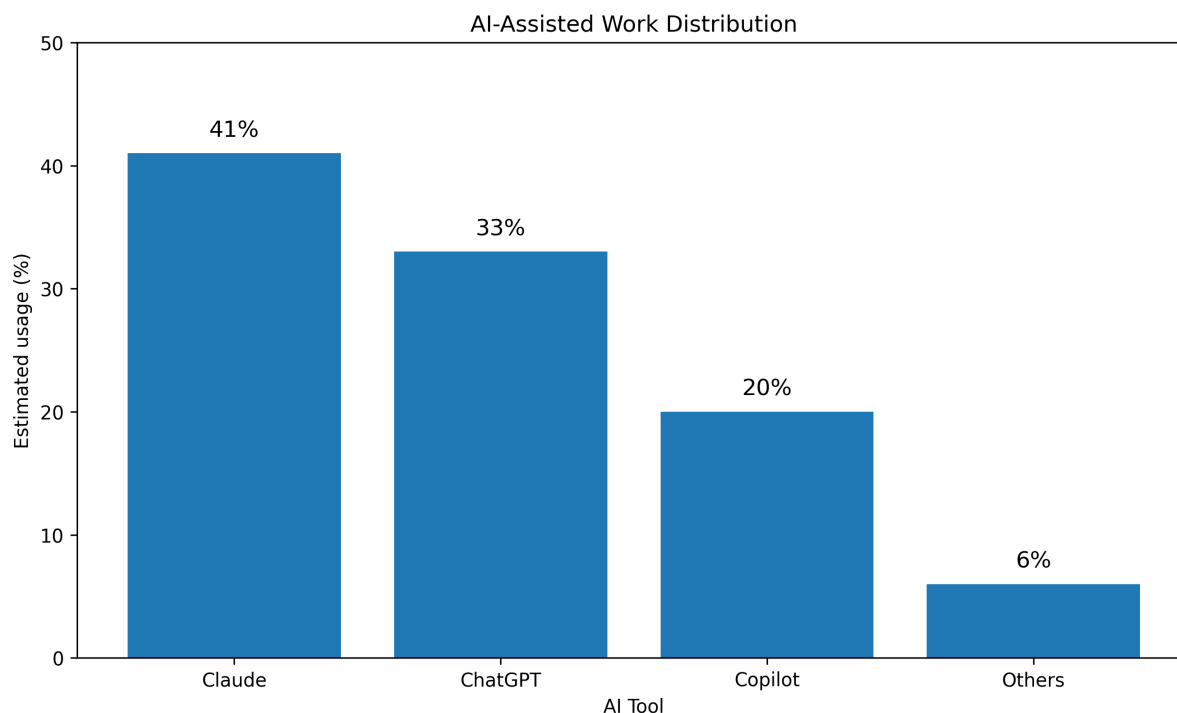


Figure 15: Reported distribution of AI-assisted work across tools

### 6.3.1 How the Data Was Obtained

At the beginning of the research, we used ChatGPT with web search to identify museums in Bangladesh and locate potentially useful sources. The search results directed us to a mixture of official museum websites, government and university pages, Wikipedia pages, online reference materials, and direct file or document links. We did not treat the AI-generated list itself as evidence. Instead, we opened the suggested sources and collected the information from the underlying pages or documents that could be checked and used for our project.

The types of information available differed from source to source. Some websites provided written information about a museum, such as its history, location, ownership, opening information,

collections, and institutional background. In other cases, the most useful evidence was available through an online gallery where individual artifacts were displayed with photographs and short descriptions. Some institutions also published journals, catalogues, articles, or scanned publications containing artifact photographs and related descriptions. We used these materials as supporting sources when they provided useful and verifiable information.

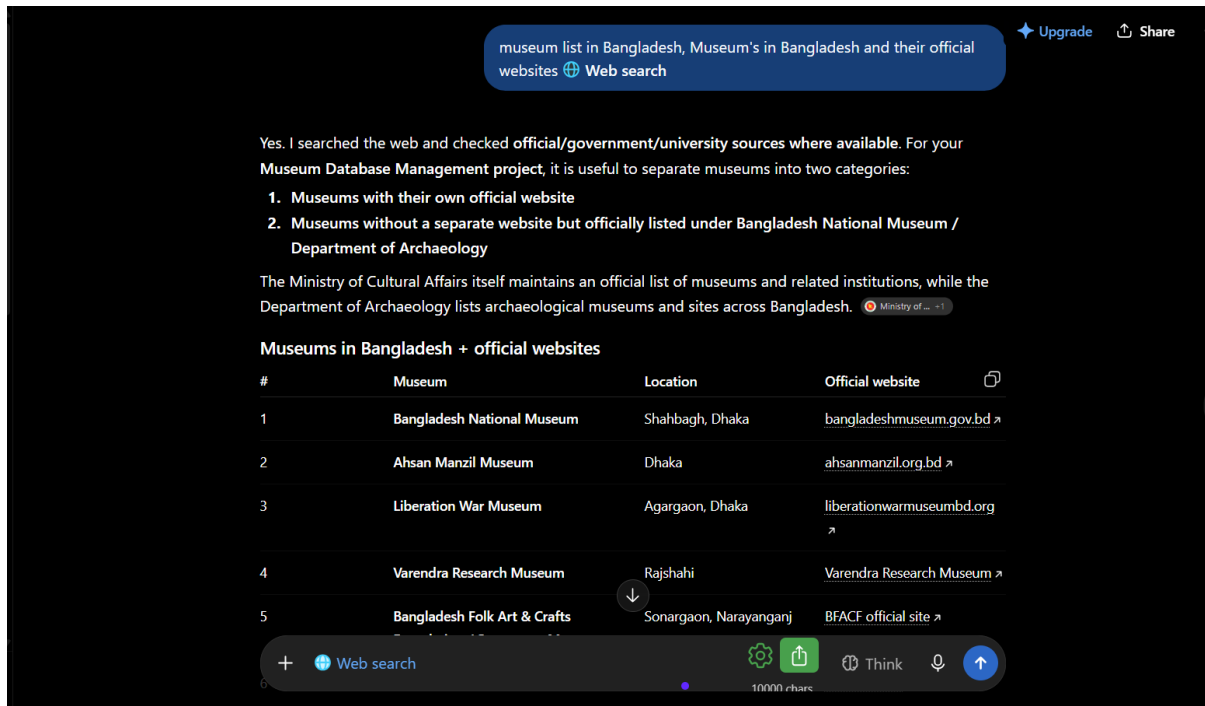


Figure 16: Example of AI-assisted web search used to identify museums and potential official or reference sources

Figure 16 provides evidence of the initial source-discovery process. The search helped us locate official websites and other relevant pages; the actual museum and artifact information was then collected from the underlying sources.

### 6.3.2 How Artifact Images Were Collected

Artifact images were collected separately from ordinary museum-level text information because reliable photographs were often available in galleries, catalogues, journals, and other publications even when a museum did not provide a complete structured artifact page. When an artifact photograph appeared in a reliable online source, we recorded the relevant source and, where permitted and appropriate for the project, captured the useful image and saved it in our shared Google Drive for later organization and reference.

In many cases, the image itself came from a journal, catalogue, or museum publication rather than from a simple webpage. During image collection, we therefore checked the surrounding publication or page so that the photograph could remain connected to the artifact information and its source. This helped us build a more useful artifact dataset instead of collecting unidentified images without context.



Figure 17: Example of artifact photographs and descriptive information collected from a museum publication or journal

Figure 17 shows the type of publication-based evidence used during artifact image collection. The photographs were useful not only as visual records but also because the accompanying text could provide historical, descriptive, or provenance information about the objects.

### 6.3.3 How Images Provided Additional Artifact Information

An important realization during the research was that artifact photographs frequently contained information that was not obvious from the artifact name alone. Museum labels visible beside or below an object could provide the artifact name, material, approximate date, finding place,

archaeological site, and other descriptive details. We therefore examined both the artifact image and any visible label rather than treating the photograph as only an illustration.

For example, the photograph of the *Krishna Killing Demon* artifact provides several useful details directly from the museum label: the artifact is identified as a stone-made object, dated to approximately the 6th century CE, and associated with the central temple at Paharpur. Information of this type helped us connect an artifact with its material, period, and archaeological finding place when preparing the structured artifact records.



Figure 18: Artifact photograph and museum label showing additional information for the Krishna Killing Demon artifact

Figure 18 demonstrates how an artifact image can function as a source of structured metadata. By reading the visible museum label, we could obtain details that supported fields such as Artifact Name, Material, Period Dating, and Finding Place.

#### 6.3.4 Source Collection

At the beginning of our data collection, we use AI-assisted deep-search prompts to discover possible museums and source pages. Some of the recurring search keywords we use are:

- museums in Bangladesh
- museum to visit in Bangladesh
- Bangladesh museums
- BNM official website
- Varendra Research Museum official website
- artifacts in Bangladesh several museums
- artifact images
- artifact list of BNM and details

The search results give us candidate links and names. We then open those links ourselves, check whether the information is useful and traceable, and record the verified sources in our shared source list. The AI output is therefore used for discovery; the original websites and documents remain the evidence for the dataset.

#### 6.3.5 Source-Based Data Extraction

After collecting a sufficient set of sources, we use AI assistance on individual sources to help locate the particular pieces of information needed for our structured tables. For artifact image work, we also search by the museum name together with an individual artifact name—for example, Vishnu avatar in BNM—and repeat the same approach for other identified artifacts and museums. When a list of artifacts is available, we check the names one by one to determine whether a usable image or additional description can be found. For museum research, we use it to help identify fields such as museum name, location, category, ownership, opening hours, entry fee, contact information, galleries, and other relevant details. For artifact research, we use it to help extract information such as artifact name, artifact type, material, period or dating, finding place, dimensions, description, donor or credit information, and available image or source references. We then compare the extracted information with the original source before entering it into our museum and artifact tables.

This approach helps us move from unstructured web pages to structured rows and columns more efficiently. It also helps us work consistently across many different sources, while the original links remain attached to the records so that the information can be traced back to its evidence.

### **6.3.6 Data Cleaning and Anomaly Checking**

We also use AI assistance during data cleaning and checking. We ask AI tools to help us identify possible duplicate records, inconsistent naming, unusual values, missing fields, null values, formatting differences, and other anomalies in the collected data. We then inspect the flagged records ourselves and return to the relevant sources when verification is required. We do not fill a missing value simply because an AI tool suggests one; when a value cannot be verified from a reliable source, we keep it missing rather than introducing unsupported information.

### **6.3.7 Data Organization and Table Preparation**

We use AI assistance to help us decide how the information collected from different sources can be organized consistently into the museum list and artifact tables. This includes suggesting suitable column names, grouping similar attributes, identifying repeated information, and checking whether a field appears to belong to the museum level or the artifact level. The final structure is decided by us after comparing the suggestions with the actual data and with the requirements of the database model.

### **6.3.8 Entity-Relationship Modeling and Documentation Support**

We also use AI as a supporting tool while reviewing possible entities, attributes, relationships, and documentation structure. However, the final Entity-Relationship model is based on our collected data and our database design decisions. We use AI to clarify database concepts, improve explanations, check wording, and organize documentation, but we review the resulting content and keep the final technical decisions under our control.

Overall, we use AI to reduce repetitive research and data-preparation effort, not to replace source verification or our own modeling decisions. This workflow allows us to benefit from AI assistance while keeping the dataset traceable and the database design grounded in the evidence we collect.

## **6.4 Lessons Learned**

The project demonstrates that data quality is a central part of database engineering. A logically correct schema is not sufficient if the underlying records are incomplete, inconsistent, or unsupported. We also learn that clear task visibility and source traceability are important when several people contribute to one shared dataset.

## **6.5 Suggestions for Improvement**

Future batches can benefit from an earlier common reference-data specification, a shared validation checklist, and a clearly defined minimum implementation milestone before extensive optional data collection begins. A common cross-group synchronization mechanism can also reduce duplicated decisions about shared geographic entities. Finally, providing an agreed artifact-source quality guideline at the beginning of the course can help teams balance coverage, reliability, and implementation time more effectively.

## 7 Logical Modelling

The relational model provides the logical structure used by a relational Database Management System to organize data as relations consisting of rows and columns. We convert the Entity-Relationship model into a relational schema by mapping entities to tables, relationships to foreign keys, attributes to columns, and identifying attributes to primary keys. The design also separates multi-valued information into dedicated relations and checks the resulting structure against normalization principles.

### 7.1 Converting Entity-Relationship Model to Relational Model

We convert the approved Entity-Relationship model into a relational model by applying a consistent set of mapping rules. The purpose of this stage is to turn the conceptual structure of the museum and artifact domain into relations that can later be implemented in MySQL Workbench. We keep the structure of the conceptual model intact while making the keys, attributes, and relationships explicit at the relational level. The mapping also ensures that information that can occur more than once for a museum or artifact is not forced into a single repeating column.

The conversion follows the main relational-mapping rules described below. We first convert entities into tables, then represent relationships through foreign keys, map attributes into columns, assign primary keys, separate multi-valued information where necessary, and finally consider normalization, integrity constraints, and indexing.

#### 7.1.1 Entities to Tables

Each strong or weak entity represented in the Entity-Relationship model becomes a relation in the logical model. The relation normally keeps the same conceptual name so that the transition from the model to the database remains easy to trace. For example, the Museum entity becomes the *Museum* relation, the Artifact entity becomes the *Artifact* relation, and the geographic entities Division, District, and City become separate relations.

This mapping gives every entity a dedicated place for its own attributes. A row in the Museum relation represents one museum, while a row in the Artifact relation represents one artifact. Similarly, a row in Gallery represents one gallery belonging to a particular museum. Separating these entities allows us to store information once and connect related records through keys instead of repeating the same descriptive information in several places.

The same rule is applied to supporting entities such as Owner, Category, Material, Artifact Type, Image, Entry Fee, Opening Hours, Closed Days, Contact, and Source. As a result, the logical model preserves the different information groups identified during conceptual modelling and prepares them for implementation as separate database tables.

#### 7.1.2 Relationships to Foreign Keys

We represent relationships from the Entity-Relationship model in the relational model mainly through foreign keys. A foreign key stores the key value of a related record and therefore connects

rows between two relations while preserving the relationship identified during conceptual modelling.

For example, the Entity-Relationship model shows that a City hosts multiple museums. Therefore, City\_ID is stored in Museum as a foreign key. In the same way, District\_ID is stored in City to represent the relationship between District and City, and Division\_ID is stored in District to represent the relationship between Division and District. Museum also stores Owner\_ID and Category\_ID to connect each museum with its owner and category.

The artifact side follows the same principle. Artifact stores Artifact\_Type\_ID and Material\_ID as foreign keys. Its relationship with Gallery is represented using Museum\_ID together with Gallery\_No because Gallery has a composite key. Through Gallery, each artifact is associated with its Museum, while the gallery number is not treated as globally unique.

For relations that represent dependent or multi-valued information, the museum identifier is also carried into the child relation. For example, Entry Fee, Opening Hours, Closed Days, Contact, and Source retain Museum\_ID so that every record can be traced back to its museum. In this way, the relational model directly reflects the connections defined in the Entity-Relationship model.

### **7.1.3 Attributes to Columns**

Each attribute in the Entity-Relationship model becomes a column in the corresponding relation. We keep attribute names sufficiently descriptive so that the meaning of a field remains clear during later SQL implementation. Key attributes become key columns, while descriptive attributes become ordinary data columns or foreign-key columns depending on their role.

For example, the Museum relation contains attributes such as Museum\_ID, Museum\_Name, Coordinates, Public\_Transit\_Access, Number\_of\_Galleries, Established\_Date, and Opened\_as\_Museum\_Date. It also contains City\_ID, Owner\_ID, and Category\_ID to connect the museum with related relations.

The Artifact relation contains Artifact\_ID, Artifact\_Name, Description, Period\_Dating, Finding\_Place, and Credit\_Donor, together with its museum, gallery, and dimension references. Artifact Type, Material, and Donor are not stored as single-valued foreign keys in Artifact because these relationships may be multi-valued. Their associations are represented through the normalized junction tables. The Gallery foreign key uses Museum\_ID and Gallery\_No because Gallery has a composite key.

This mapping allows the logical model to preserve the detailed information collected during the data-acquisition stage while keeping the database structure organized and suitable for structured queries.

### **7.1.4 Primary Keys**

We assign a primary key to every relation so that each stored record can be uniquely identified. A primary key may consist of one attribute when that attribute is sufficient to identify a record, or it may consist of more than one attribute when the identity of a record depends on a combination of values.

The single-attribute primary keys in our current logical model are shown below. We use underlining to follow the standard relational notation for identifying primary-key attributes.

Table 22: Single-attribute primary keys in the relational model

| <b>Relation</b> | <b>Primary Key</b>      |
|-----------------|-------------------------|
| Division        | <u>Division_ID</u>      |
| District        | <u>District_ID</u>      |
| City            | <u>City_ID</u>          |
| Owner           | <u>Owner_ID</u>         |
| Category        | <u>Category_ID</u>      |
| Museum          | <u>Museum_ID</u>        |
| Artifact Type   | <u>Artifact_Type_ID</u> |
| Material        | <u>Material_ID</u>      |
| Artifact        | <u>Artifact_ID</u>      |

The remaining relations require composite primary keys because one attribute alone does not provide sufficient uniqueness. We use the following combinations:

Table 23: Composite primary keys in the relational model

| <b>Relation</b> | <b>Composite Primary Key</b>              |
|-----------------|-------------------------------------------|
| Gallery         | <u>Museum_ID</u> , <u>Gallery_No</u>      |
| Image           | <u>Artifact_ID</u> , <u>Image_URL</u>     |
| Entry Fee       | <u>Museum_ID</u> , <u>Fee_Type</u>        |
| Source          | <u>Museum_ID</u> , <u>Source_Link</u>     |
| Opening Hours   | <u>Museum_ID</u> , <u>Day_Name</u>        |
| Closed Days     | <u>Museum_ID</u> , <u>Closed_Day_Name</u> |

The use of a composite key is important in these relations. For Gallery, for example, Gallery\_No alone cannot uniquely identify a gallery because different museums can use the same gallery number. The pair (Museum\_ID, Gallery\_No) identifies one gallery within one museum. Similarly, the same fee type, source URL, day name, or closed-day name may occur for different museums, so Museum\_ID is retained as part of the key. For Image, Artifact\_ID is combined with Image\_URL because an artifact may have more than one image and the image URL distinguishes the individual image record.

This explicit separation between single and composite primary keys makes the relational design easier to implement and also makes the uniqueness rules visible before database creation begins.

### 7.1.5 Normalization

Normalization was derived from the actual structure of the museum and artifact dataset. The purpose was not simply to increase the number of tables, but to remove repeating groups,

separate facts according to what determines them, and preserve relationships that occur more than once. The final design therefore follows a data-driven sequence from a wide collection view to atomic relations and then to a normalized relational schema.

### Step 0 — Identify the repeating and multi-valued facts.

During collection, museum and artifact information was naturally gathered in wide working records. This is useful for research, but it is not an appropriate final relational structure when one artifact has several independent values. The loaded dataset demonstrates that some artifacts are associated with multiple donors, multiple artifact types, and multiple materials. A museum can also have multiple telephone numbers. If such values were stored as comma-separated text inside one row, they would represent lists rather than separate relational facts and could not be independently keyed or referenced.

Table 24: From wide collection data to atomic relational facts

| Wide / repeating representation                      | Why it is problematic                                                                  | Final relational representation      |
|------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------|
| One Artifact row containing several donor names      | Donor values form a repeating group and are difficult to validate or join individually | donor +<br>artifact_has_donor        |
| One Artifact row containing several artifact types   | Multiple classifications are packed into one attribute                                 | artifact_type +<br>artifact_has_type |
| One Artifact row containing several materials        | Independent material facts are mixed into one field                                    | material +<br>artifact_has_material  |
| One Artifact row containing several image references | Image records cannot be independently maintained                                       | image                                |
| One Museum row containing several phone numbers      | Phone values repeat inside one attribute and cannot be individually keyed              | museum_phone                         |
| Museum rows repeating geographic names               | The same district/division facts are repeated across museums                           | division →district<br>→city →museum  |

### Step 1 — First Normal Form (1NF): make repeating facts atomic.

The first normalization decision was to ensure that one field represents one value for one record rather than an embedded list. The Artifact relation therefore does not store several donors, artifact types, or materials in a single comma-separated attribute. The same principle is applied to museum telephone numbers and artifact images. Each independent association is represented by a separate row that can be identified, referenced, queried, and validated.

This is supported by the actual loaded data: 111 artifacts are associated with multiple donors, 571 artifacts with multiple artifact types, and 427 artifacts with multiple materials. These observations demonstrate why a single-valued donor/type/material attribute would not faithfully represent the collected dataset.

**Step 2 — Second Normal Form (2NF): remove partial dependencies from composite-key relationships.**

The M:N relationships require composite keys in their relational implementation. To demonstrate the dependency issue, consider an incorrect relationship table of the form

(Artifact\_ID, Donor\_ID, Artifact\_Name, Donor\_Name)

Here, `Artifact_Name` depends only on `Artifact_ID`, while `Donor_Name` depends only on `Donor_ID`. Because neither non-key attribute depends on the whole composite key, the design contains partial dependencies.

The final design removes these partial dependencies. Artifact facts are stored in `artifact`, donor facts are stored in `donor`, and `artifact_has_donor` contains only the identifiers required to represent the association. The same decomposition is used for Artifact–Artifact Type and Artifact–Material.

Table 25: 2NF demonstration for the M:N relationships

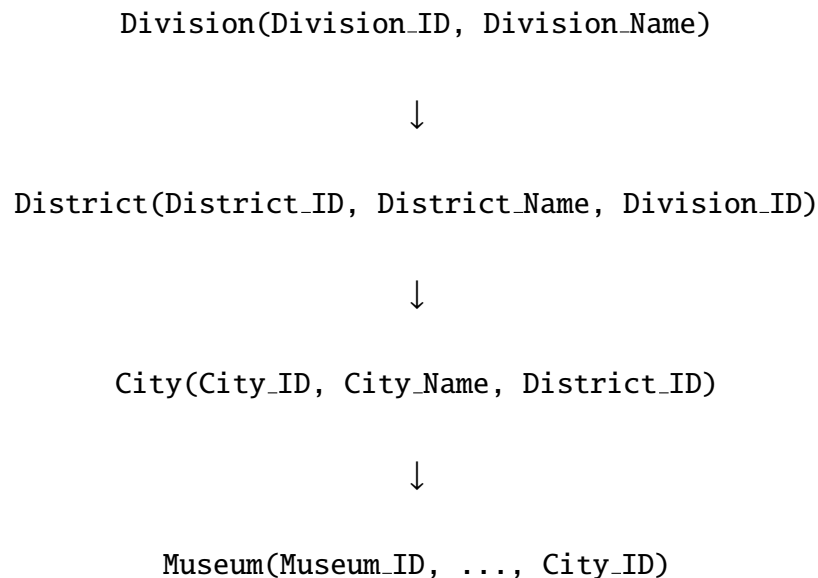
| Design stage                    | Dependency problem                                                                                                                                                                               | Normalized solution                                                                                                                                |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Incorrect combined relationship | <code>Artifact_Name</code> depends on <code>Artifact_ID</code> ; <code>Donor_Name</code> depends on <code>Donor_ID</code> . These attributes therefore depend on only part of the composite key. | Descriptive attributes are moved to their parent relations: <code>artifact</code> stores artifact facts and <code>donor</code> stores donor facts. |
| Artifact–Donor                  | Composite key = ( <code>Artifact_ID</code> , <code>Donor_ID</code> ). Artifact and donor descriptions do not depend on the complete pair.                                                        | <code>artifact_has_donor</code><br>( <code>Artifact_ID</code> , <code>Donor_ID</code> )                                                            |
| Artifact–Type                   | Composite key = ( <code>Artifact_ID</code> , <code>Artifact_Type_ID</code> ). Type descriptions would repeat if stored in the relationship rows.                                                 | <code>artifact_has_type</code><br>( <code>Artifact_ID</code> ,<br><code>Artifact_Type_ID</code> )                                                  |
| Artifact–Material               | Composite key = ( <code>Artifact_ID</code> , <code>Material_ID</code> ). Material descriptions would repeat if stored in the relationship rows.                                                  | <code>artifact_has_material</code><br>( <code>Artifact_ID</code> ,<br><code>Material_ID</code> )                                                   |

**Step 3 — Third Normal Form (3NF): separate facts according to their determinants.**

The geographic hierarchy provides a clear example. A museum belongs to a city; a city belongs to a district; and a district belongs to a division. If Museum stored `City_Name`, `District_Name`, and `Division_Name` directly, the same geographic facts would be repeated across many museum

rows. The dependency chain can be represented as  $\text{City} \rightarrow \text{District}$  and  $\text{District} \rightarrow \text{Division}$ , rather than storing all higher-level names in every Museum row.

The final decomposition is:



The same principle is applied to independent master data. `Owner`, `Category`, `Donor`, `Artifact_Type`, and `Material` each keep their descriptive names in their own relations and are referenced through identifiers. This prevents the same descriptive fact from being copied into many unrelated rows.

#### **Step 4 — Normalize other dependent and multi-valued relations.**

Artifact-specific images are represented by `image`, so one artifact can have multiple image records without repeating all artifact attributes. Museum telephone numbers are represented by `museum_phone`. Gallery uses the composite key (`Museum_ID`, `Gallery_No`) because a gallery number is meaningful within a museum rather than globally. `Entry Fee`, `Opening Hours`, `Closed Days`, `Contact`, and `Source` are also separated from `Museum` so that their records can be maintained and queried independently.

#### **Step 5 — Enforce normalization with primary and foreign keys.**

The normalization is reflected in the implemented MySQL schema. Primary keys identify individual records; foreign keys preserve the relationships between parent and dependent relations; and composite primary keys prevent duplicate relationship rows. For example, `artifact_has_donor` uses (`Artifact_ID`, `Donor_ID`), `artifact_has_type` uses (`Artifact_ID`, `Artifact_Type_ID`), and `artifact_has_material` uses (`Artifact_ID`, `Material_ID`). Gallery uses (`Museum_ID`, `Gallery_No`), which also reflects the museum-specific meaning of `Gallery_No`.

#### **Step 6 — Demonstrate that the normalization is required by the actual dataset.**

The following counts were obtained from the loaded database and provide evidence that the multi-valued relationships are real characteristics of the dataset rather than assumptions introduced only for modelling purposes.

Table 26: Observed evidence supporting the normalized design

| Observed condition                                | Count | Database evidence     |
|---------------------------------------------------|-------|-----------------------|
| Artifacts associated with multiple donors         | 111   | artifact_has_donor    |
| Artifacts associated with multiple artifact types | 571   | artifact_has_type     |
| Artifacts associated with multiple materials      | 427   | artifact_has_material |
| Museums with multiple phone records               | 3     | museum_phone          |

### Step 7 — Compare the alternative with the final normalized model.

Table 27: Unnormalized versus normalized museum database design

| Aspect                     | Wide / unnormalized approach                                                   | Final normalized approach                                                |
|----------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Multi-valued artifact data | Multiple donors, types, or materials packed into Artifact                      | Separate master relations plus M:N junction tables                       |
| Update anomaly             | Repeated or embedded values may need multiple edits                            | The master fact is stored once and relationship rows point to it         |
| Insertion                  | A new relationship may require editing an existing list                        | A parent record and a new relationship row can be inserted independently |
| Deletion                   | Removing an association can remove the only stored copy of a descriptive value | Descriptive master data and relationship rows are separated              |
| Referential integrity      | Embedded text lists cannot be enforced as foreign keys                         | PK/FK constraints enforce the implemented relationships                  |
| Query flexibility          | Parsing embedded lists is required                                             | SQL JOIN, GROUP BY, and aggregation directly traverse relationships      |
| Scalability                | Repeating groups grow inside individual rows                                   | New associations are represented by additional relationship rows         |

### Conceptual-to-relational distinction.

The conceptual Chen ER diagram represents Artifact–Donor, Artifact–Artifact Type, and Artifact–Material as M:N relationships between entities. The junction tables are introduced only during relational implementation. They are therefore not treated as new conceptual entities. This distinction is important: the ER model communicates the business relationship, while the relational schema implements that relationship using keys and foreign keys.

### Normalization conclusion.

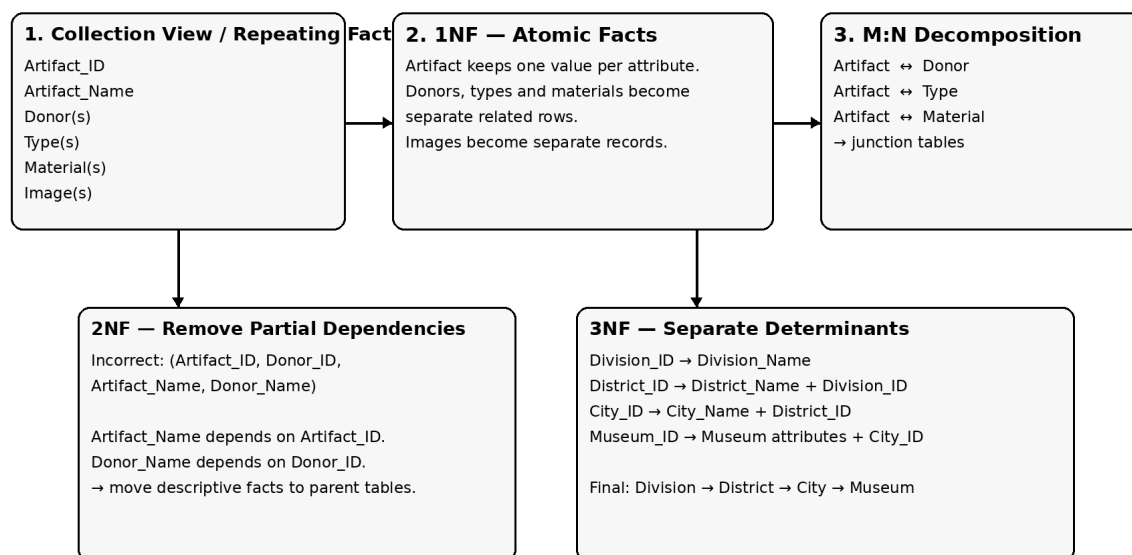
The final model uses normalization as a direct response to the structure of the collected data. 1NF addresses repeating and multi-valued facts by making them atomic; 2NF removes partial dependencies from composite-key relationship structures; and 3NF separates facts according

to their determinants, particularly in the geographic and master-data hierarchy. The result is a schema in which independent facts are stored once, M:N associations are explicit, and the database can enforce and query those relationships without relying on comma-separated lists or duplicated descriptive values.

### 7.1.6 Worked Normalization Demonstration

The normalization can be demonstrated as a sequence of concrete modelling decisions rather than as a list of definitions. Figure 19 summarizes the progression from a wide collection view to atomic relations, then to M:N junction tables and the geographic dependency chain. The figure is a modelling demonstration derived from the actual structure of the project dataset; it does not introduce an additional source dataset.

#### Worked Normalization Demonstration — Museum Artifact Data



Evidence in loaded dataset: 111 multi-donor | 571 multi-type | 427 multi-material | 3 museums with multiple phone records

Figure 19: Step-by-step normalization demonstration for the museum artifact data

**Worked example A — Artifact and multiple classifications.** Consider an artifact record that is associated with more than one artifact type. In the loaded database, the SQL evidence shows artifacts with three distinct artifact types; for example, the displayed result includes Artifact ID 16 (Mahamaya) with three types. A single attribute such as `Artifact_Type` cannot represent that situation without either repeating columns or storing a delimited list. The normalized solution stores the artifact once in `artifact`, stores each type definition once in `artifact_type`, and stores each association as a row in `artifact_has_type`.

**Worked example B — 2NF on a relationship table.** Suppose the relationship between an artifact and a donor were stored as:

`R(Artifact_ID, Donor_ID, Artifact_Name, Donor_Name)`

with composite key (`Artifact_ID`, `Donor_ID`). The functional dependencies are:

$$\begin{aligned}\text{Artifact\_ID} &\rightarrow \text{Artifact\_Name} \\ \text{Donor\_ID} &\rightarrow \text{Donor\_Name} \\ (\text{Artifact\_ID}, \text{Donor\_ID}) &\rightarrow \text{association}\end{aligned}$$

The first two dependencies are partial dependencies because each descriptive attribute depends on only part of the composite key. Therefore, the relation is not in 2NF. Decomposing it into `artifact`, `donor`, and `artifact_has_donor` removes those partial dependencies. The same reasoning applies to the `Artifact-Type` and `Artifact-Material` relationships.

**Worked example C — 3NF in the geographic hierarchy.** The museum relation does not need to repeat the names of its city, district, and division. The dependency chain represented by the final schema is:

$$\begin{aligned}\text{Museum\_ID} &\rightarrow \text{City\_ID} \\ \text{City\_ID} &\rightarrow \text{District\_ID} \\ \text{District\_ID} &\rightarrow \text{Division\_ID}\end{aligned}$$

If the higher-level geographic names were copied into every museum row, those values would be transitively dependent on the museum identifier through the intermediate geographic identifiers. The decomposition into `division`, `district`, `city`, and `museum` keeps each geographic fact with its determinant and avoids repeated district/division descriptions.

**Worked example D — Why the junction tables are not conceptual entities.** At the conceptual level, the business fact is “an artifact can be associated with many donors, and a donor can be associated with many artifacts.” This is represented as an M:N relationship in the Chen ER diagram. During relational implementation, the M:N relationship requires a relation whose key contains both participating identifiers. Hence `artifact_has_donor(Artifact_ID, Donor_ID)` is an implementation mechanism for the conceptual relationship, not a new business entity. The same distinction is maintained for `artifact_has_type` and `artifact_has_material`.

**Worked example E — Database evidence closes the modelling argument.** The need for these decompositions is observable in the loaded data: 111 artifacts have multiple donor associations, 571 have multiple artifact-type associations, and 427 have multiple material associations. In addition, 3 museums have multiple phone records. These are not hypothetical relationship requirements; they are properties of the loaded dataset that the relational model must represent without repeating groups.

Table 28: Normalization decision trail: problem, dependency, decomposition, and evidence

| Observed modelling problem           | Dependency / rule                                             | Relational decision                                                | Evidence                               |
|--------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------|
| Multiple donors for one artifact     | Artifact and donor are independently identified               | <code>artifact + donor + artifact_has_donor</code>                 | 111 multi-donor artifacts              |
| Multiple artifact types              | One artifact may participate in several type associations     | <code>artifact_type + artifact_has_type</code>                     | 571 multi-type artifacts               |
| Multiple materials                   | One artifact may participate in several material associations | <code>material + artifact_has_material</code>                      | 427 multi-material artifacts           |
| Multiple phones per museum           | Phone number is a repeating museum-level fact                 | <code>museum_phone</code>                                          | 3 museums with multiple phones         |
| Geographic repetition                | City determines District; District determines Division        | <code>city + district + division</code>                            | 52 cities / 32 districts / 8 divisions |
| Gallery number reused across museums | Gallery number is meaningful within its museum                | Composite key ( <code>Museum_ID</code> , <code>Gallery_No</code> ) | 338 gallery records                    |

This worked sequence makes the normalization auditable: each decomposition is tied to an observed data characteristic, a dependency or key rule, and a concrete relation in the implemented schema. The result is not normalization for its own sake; it is a direct translation of the data's actual cardinalities and dependencies into a relational structure.

### 7.1.7 Constraints

Primary-key constraints enforce uniqueness and prevent a relation from containing two rows with the same primary-key value. Foreign-key constraints preserve the relationships represented in the Entity-Relationship model and prevent a child record from referring to a non-existing parent record.

We also identify the need for appropriate data-type, null ability, and value constraints during implementation. For example, identifier fields should use suitable integer or string types according to the final data definition, while time, date, and numeric fields should use data types that support the intended queries. The implementation constraints are defined in the MySQL schema and exercised by the ETL and validation workflow.

These constraints are reflected in the implemented MySQL schema and were checked through the ETL pre-flight and post-load validation workflow.

### 7.1.8 Indexes

Primary keys and foreign keys provide the core indexes required for the implemented schema; additional indexing can be evaluated as the SQL workload grows. Primary keys will naturally

require indexing, and foreign-key columns will also be considered for indexing because they are frequently involved in joins between related relations.

Additional indexes can also be evaluated for columns that are frequently used in filtering, joining, grouping, and location-based retrieval. For example, museum location attributes and foreign keys connecting Museum with City, Category, Owner, and Artifact-related relations may become important for tourism-oriented queries. Rather than adding indexes without evidence, we base any additional indexing decisions on the actual query workload and performance observed during testing.

Overall, this conversion gives us a clear path from the conceptual Entity-Relationship model to a relational structure that can be implemented consistently. Each entity has a corresponding relation, each relationship has an explicit key-based representation, attributes are mapped to meaningful columns, and primary and composite keys make record identity unambiguous. The resulting design provides the logical foundation for the implemented MySQL Workbench database.

## 7.2 Relational Schema for Museum and Artifact Data Integration

The logical schema preserves the geographic hierarchy and separates museum and artifact concerns while maintaining the links needed for tourism-oriented queries. It also preserves source provenance so that museum information can be traced back to the references used during collection. The current relational implementation contains 22 relations, including junction tables for the many-to-many artifact relationships.

Table 29: Relational schema implemented in MySQL

| Relation              | Key and important attributes                                                                                                                                                 |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Division              | Division_ID (PK), Division_Name                                                                                                                                              |
| District              | District_ID (PK), District_Name, Division_ID (FK)                                                                                                                            |
| City                  | City_ID (PK), City_Name, District_ID (FK)                                                                                                                                    |
| Owner                 | Owner_ID (PK), Owner_Name                                                                                                                                                    |
| Category              | Category_ID (PK), Category_Name                                                                                                                                              |
| Museum                | Museum_ID (PK), Museum_Name, City_ID (FK), Owner_ID (FK), Category_ID (FK), Coordinates, Public_Transit_Access, Number_of_Galleries, Established_Date, Opened_as_Museum_Date |
| Gallery               | Museum_ID (PK, FK), Gallery_No (PK), Gallery_Name, Floor                                                                                                                     |
| Dimension             | Dimension_ID (PK), Length, Width, Height                                                                                                                                     |
| Artifact              | Artifact_ID (PK), Artifact_Name, Museum_ID (FK), Gallery_No (FK), Dimension_ID (FK), Period_Dating, Finding_Place, Provenance_Note, Description, Source_Link                 |
| Donor                 | Donor_ID (PK), Donor_Name                                                                                                                                                    |
| Artifact Type         | Artifact_Type_ID (PK), Artifact_Type_Name                                                                                                                                    |
| Material              | Material_ID (PK), Material_Name                                                                                                                                              |
| Artifact Has Donor    | Artifact_ID (PK, FK), Donor_ID (PK, FK)                                                                                                                                      |
| Artifact Has Type     | Artifact_ID (PK, FK), Artifact_Type_ID (PK, FK)                                                                                                                              |
| Artifact Has Material | Artifact_ID (PK, FK), Material_ID (PK, FK)                                                                                                                                   |
| Image                 | Image_ID (PK), Artifact_ID (FK), Image_URL, Source_Link                                                                                                                      |
| Museum Phone          | Museum_ID (PK, FK), Phone_Number (PK)                                                                                                                                        |
| Entry Fee             | Museum_ID (PK, FK), Fee_Type (PK), Fee_Amount                                                                                                                                |
| Opening Hours         | Museum_ID (PK, FK), Day_Name (PK), Opening_Time, Closing_Time                                                                                                                |
| Closed Days           | Museum_ID (PK, FK), Closed_Day_Name (PK)                                                                                                                                     |
| Contact               | Museum_ID (PK, FK), Website, Email                                                                                                                                           |
| Source                | Museum_ID (PK, FK), Source_Link (PK)                                                                                                                                         |

Table 29 documents the 22 relations implemented in MySQL. The M:N relationships are represented conceptually as relationships in the Chen ER diagram and are implemented relationally by the three junction tables. This separation keeps the conceptual model readable while providing a normalized database structure for SQL queries.

### 7.3 Schema Design Rationale

The geographic relations are separated so that Division, District, and City values are not repeatedly stored as free text in every museum record. Museum-specific descriptive information is kept in Museum, while Owner and Category are separated to avoid repeating the same descriptive value across many museums. Gallery uses a composite key because a gallery number is meaningful within a particular museum rather than globally.

Artifact is connected to its Gallery and Museum while its multi-valued donor, artifact-type, and material associations are represented through junction tables. This structure allows artifact queries to combine collection-level information with descriptive characteristics without repeating values inside Artifact. Image is separated because an artifact can have more than one available image and because each image may have its own source link.

Entry Fee, Opening Hours, and Closed Days are also separated because these properties can be multi-valued. A museum can have different fee types and different opening or closing arrangements across days. Contact is kept as a separate relation because website, phone, and email form a distinct contact-information group. Source is kept explicitly to preserve provenance and support future verification and maintenance.

## 8 Course Pedagogy Summary

The project gave us experience that is different from a conventional database exercise. Instead of receiving a small, predefined dataset, we worked with real and heterogeneous information, interpreted an open-ended problem, coordinated tasks through Scrum-inspired planning, and prepared data for integration. The following summary captures the most practical differences, lessons, and improvements we identified during the project.

### 8.1 Comparison of Course Pedagogy

Table 30: Comparison of conventional course work and the Database Systems project

| Aspect                    | Traditional Course Approach                                    | Database System Project Approach                                                                                                   |
|---------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data</b>               | Mostly predefined and structured datasets.                     | Real-world and heterogeneous data collected from multiple sources, requiring source checking and preparation.                      |
| <b>Problem Definition</b> | Usually clearly provided by the instructor.                    | Requires research, interpretation of requirements, and decisions about what information should be represented.                     |
| <b>Teamwork</b>           | Often limited to small group activities.                       | Continuous collaboration within the group and coordination with other groups through the wider Scrum structure.                    |
| <b>Data Processing</b>    | Limited practical data cleaning and transformation.            | Extensive collection, cleaning, standardization, validation, source tracking, and preparation for ETL.                             |
| <b>Project Management</b> | Less emphasis on formal task management.                       | Uses Scrum-inspired sprint planning, task distribution, Trello tracking, and progress monitoring.                                  |
| <b>Integration</b>        | Usually focuses on an individual solution or isolated dataset. | Requires integration of datasets, common geographic references, ER models, and sector-level outputs.                               |
| <b>Industry Exposure</b>  | Mostly theoretical or laboratory-based.                        | Provides an industry-like experience through real data, source provenance, collaboration, validation, and database implementation. |

## 8.2 Major Lessons Learned from the Project

Table 31: Major lessons learned during the project

| Area                           | Lesson Learned                                                                                                                                                                                                               |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Industry Vibe</b>           | Real-world data are often incomplete, inconsistent, and distributed across multiple sources. Reliable database work therefore requires continuous investigation and validation rather than simply entering available values. |
| <b>Collaboration</b>           | Large projects require clear task distribution, regular communication, shared resources, and coordination so that corrections made by one member are reflected in the common dataset.                                        |
| <b>Conflict Resolution</b>     | Technical disagreements can be handled through discussion, source verification, comparison of alternatives, and consensus-based decisions rather than assumptions.                                                           |
| <b>Project Management</b>      | Sprint-based planning and clear task priorities help us divide a large research problem into manageable work and monitor what remains unfinished.                                                                            |
| <b>Data Management</b>         | Cleaning, transformation, standardization, validation, and provenance checking are essential before a database can be considered reliable.                                                                                   |
| <b>Documentation</b>           | Proper documentation and source tracking make the work reproducible and make later corrections easier when external information changes.                                                                                     |
| <b>AI-Assisted Development</b> | AI tools can accelerate repetitive research and preparation, but human verification, source checking, and domain understanding remain essential for final decisions.                                                         |

## 8.3 Suggestions for Improving the Course Pedagogy

Table 32: Suggestions for improving the project-based course pedagogy

| Area                            | Suggestion                                                                                                                                                                         |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Integration Guideline</b>    | Provide a common integration contract or database-modelling guideline before the first sprint so that groups use compatible identifiers and naming conventions from the beginning. |
| <b>ETL Training</b>             | Provide a short practical demonstration of data extraction, transformation, validation, and loading before students begin working with heterogeneous real-world datasets.          |
| <b>Sample Dataset</b>           | Provide a small pre-cleaned dataset so that students can understand the expected data-processing workflow before starting large-scale source collection.                           |
| <b>Data Collection Time</b>     | Allocate sufficient time for collection, cleaning, and validation because real-world data require considerable manual investigation and cross-checking.                            |
| <b>Inter-Group Coordination</b> | Arrange periodic coordination sessions among Scrum levels so that shared entities, naming conventions, and integration issues can be resolved earlier.                             |
| <b>Version Control</b>          | Provide basic Git/GitHub workflow guidance so that datasets, ETL scripts, schema files, and documentation can be maintained systematically.                                        |
| <b>Evaluation</b>               | Evaluate both the final database output and the development process, including data quality, documentation, collaboration, project management, and validation.                     |

This summary complements the earlier discussion of opportunities, limitations, and lessons learned without repeating those sections verbatim. It also connects the course experience directly to the practical workflow we followed in source collection, ETL preparation, modelling, collaboration, and implemented database workflow.

## 9 Conclusion and Future Work

### 9.1 Conclusion

Bangladesh hosts numerous museums holding archaeological, historical, artistic, scientific, and cultural materials, yet detailed information about these institutions and their collections remains scattered across heterogeneous sources. Group-11 addresses this problem by organizing museum-level and artifact-level information for the Tourism (Museum) sector into a structured relational database design.

We identify museums across Bangladesh, collect museum-level information, investigate

museum-specific artifact information where sufficiently detailed public sources are available, verifies the collected records, and derives an Entity-Relationship model from the structure of the real data. The current dataset contains 63 museum records and 3,505 artifact records. Artifact records currently occur for 34 of the 63 museums, while 89 sources are represented in the broader research source analysis, while 74 source rows are loaded into the current relational source table. The normalized implementation separates these records across subject-specific and relationship tables. The resulting relational implementation contains 22 tables/relations covering geographic information, museums, galleries, artifacts, donors, artifact classifications, materials, dimensions, images, museum phones, visiting information, contacts, owners, categories, and source provenance. Many-to-many artifact relationships are implemented through dedicated junction tables.

The significance of the project lies in bringing fragmented museum information into a form that can support structured retrieval for tourism. A future queryable implementation can help tourists identify museums by location and visiting conditions, while the detailed artifact structure can also support museum visitors, students, researchers, and heritage-related work. The explicit source relation further supports traceability and future maintenance.

The current stage provides a structured and validated implementation: the data-discovery and modelling work has been followed by ETL loading into MySQL and SQL-based validation. The project therefore demonstrates the path from heterogeneous source data through normalization and relational implementation to queryable results.

## **9.2 Limitations**

The current implementation is a working relational database loaded and validated in MySQL Workbench. Performance benchmarking and a user-facing application remain future activities. Artifact-level coverage is also incomplete because detailed, structured artifact information is not publicly available for every museum. Museum-specific artifact tables are available for a subset of museums where enough public information exists, while other museums currently have mainly museum-level records. Even within the available artifact tables, fields such as artifact images, precise dimensions, period information, and finding places can remain unavailable.

External source information can also change over time. Museum opening hours, entry fees, websites, contact information, and other details may require periodic verification. The project therefore requires continuing source-based maintenance if it is extended into a long-term information service.

## **9.3 Future Work**

The immediate next steps are to maintain source links, periodically re-run the ETL validation workflow after verified dataset updates, and extend artifact-level coverage where reliable public information becomes available.

After loading, we developed and tested retrieval, filtering, join, and aggregation queries representing tourist, visitor, student, researcher, and administrative use cases. Example query areas include finding museums by city or district, retrieving artifacts by material or period,

comparing entry fees, checking opening information, counting museums by geographic area, and identifying museums that contain artifacts of a particular type.

We also plan to extend artifact-level coverage to additional museums as reliable public sources become available. Missing values can be reduced by revisiting source records and adding newly verified information and images. A later stage can provide a user-facing web or mobile interface so that tourists can explore museum locations, opening information, entry fees, galleries, and artifacts more conveniently.

Once the database becomes sufficiently complete and validated, richer analytics and recommendation features can also be explored. Such features can support tourism planning, museum discovery, and cultural heritage exploration without changing the core relational foundation of the project.

## 10 Bibliography

The report uses the our maintained dataset and source records as its primary project evidence. The museum and artifact information remains traceable through the shared spreadsheet links included in Table 5 and Table 5. The Entity-Relationship design is based on the project's collected data and the final conceptual model included in the documentation.

1. Group-11 (Minimalist), *Bangladesh Tourism Museum Project Dataset*, shared Google Sheets workbook containing museum, artifact, source, and museum-specific research sheets.
2. *Database Systems Project Report Template 2026*, documentation template supplied for the Database Systems Lab project.